Insurance Policy Administration System (PAS)

Business Requirement Document (BRD)

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1. Introduction

1.1 Project Overview

The **Insurance Policy Administration System (PAS)** is a web-based enterprise application designed to automate the complete lifecycle of an insurance policy—from customer onboarding to policy closure. The system enables insurance companies to efficiently manage customer registration, KYC verification, product selection, quotation, underwriting, policy issuance, premium collection, claim processing, and policy renewal.

The application supports multiple insurance products such as **Life Insurance, Health Insurance, Motor Insurance, Travel Insurance, Property Insurance, and Term Insurance**. It provides role-based access, secure document management, automated workflows, real-time reporting, and integration with external services such as payment gateways, SMS/Email notifications, and third-party verification systems.

The solution is developed using **Core Java, JDK 8, Spring Boot, Spring MVC, Spring Data JPA, Hibernate (ORM), HQL, JPQL, RESTful APIs, MySQL, Angular, Bootstrap, HTML5, CSS3, and JWT Authentication**.

1.2 Business Objectives

The primary objectives of the system are:

- Automate the insurance policy lifecycle.
- Reduce manual processing time.
- Improve customer experience.
- Provide online policy purchase and renewal.
- Automate premium calculations.
- Improve underwriting accuracy.
- Enable online claim registration and settlement.

- Ensure regulatory compliance.
- Maintain complete audit logs.
- Generate business reports and dashboards.

1.3 Project Scope

The application covers the complete insurance lifecycle.

In Scope

- Customer Registration
- KYC Verification
- Insurance Product Management
- Quote Generation
- Premium Calculation
- Underwriting
- Policy Issuance
- Premium Schedule
- Premium Collection
- Policy Renewal
- Claim Management
- Investigation
- Claim Settlement
- Policy Closure
- Reports & Dashboard
- Notification Services
- Document Management

Out of Scope

- Third-party banking core systems

- Mobile Application (Phase-2)
- AI-based Fraud Detection (Future Enhancement)
- International Insurance Regulations

1.4 Business Workflow

Customer Registration



KYC Verification



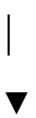
Insurance Product Selection



Quote Generation



Premium Calculation

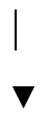


Underwriting



Policy Approval

Policy Issuance



Premium Schedule Creation



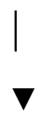
Premium Payment



Policy Active



Claim Registration



Claim Investigation



Claim Settlement



Policy Renewal / Policy Closure

1.5 System Architecture

Angular UI

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REST API (Spring Boot)

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Service Layer

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Repository Layer (JPA/Hibernate)

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MySQL Database

1.6 Technology Stack

Backend

Technology	**Version**
Core Java	JDK 8
Spring Boot	3.x
Spring MVC	Latest
Spring Data JPA	Latest
Hibernate ORM	Latest
HQL	Hibernate Query Language
JPQL	Java Persistence Query Language
RESTful API	JSON
Maven	Build Tool

Frontend

Technology	Version
Angular	18+
Bootstrap	5
HTML5	Latest
CSS3	Latest
TypeScript	Latest
JavaScript	ES6

Database

Technology	Version
MySQL	8.x

Security

- Spring Security
- JWT Authentication
- BCrypt Password Encryption
- Role-Based Access Control (RBAC)
- Session Management
- Audit Logging

1.7 User Roles

Role	Responsibilities
Customer	Register, Buy Policy, Pay Premium, Raise Claim
Agent	Customer Onboarding, Quote Generation, Policy Selling
Underwriter	Risk Assessment, Policy Approval
Claim Officer	Claim Verification and Settlement
Finance Officer	Premium Collection and Refund
Administrator	Master Configuration, User Management, Reports

1.8 Functional Modules

Module No	Module Name
1	Customer Registration
2	KYC Verification
3	Insurance Product Management
4	Quote Generation
5	Underwriting
6	Policy Issuance
7	Premium Schedule
8	Premium Payment
9	Policy Servicing
10	Claim Management
11	Investigation
12	Claim Settlement

Module No	Module Name
13	Policy Renewal
14	Policy Closure
15	Reports & Dashboard

1.9 Non-Functional Requirements

Performance

- Response time less than **3 seconds**.
- Support **5,000+ concurrent users**.
- Handle **100,000+ policies**.

Security

- JWT Token Authentication
- HTTPS Communication
- Password Encryption
- Role-Based Authorization
- Audit Trail

Availability

- System Availability: **99.9%**
- Daily Database Backup
- Disaster Recovery Support

Scalability

- Horizontal Scaling

- Cloud Deployment Ready
- Load Balancer Support
- Microservice Compatible

1.10 Master Modules

The following master modules are maintained by the Administrator.

- Department Master
- Designation Master
- User Master
- Role Master
- Insurance Product Master
- Policy Type Master
- Risk Category Master
- Coverage Master
- Rider Master
- Premium Frequency Master
- Payment Mode Master
- Claim Type Master
- Document Master
- Country Master
- State Master
- City Master
- Branch Master
- Tax Master

1.11 Integration

The application integrates with:

- Payment Gateway
- SMS Gateway
- Email Service
- Aadhaar/PAN Verification (if applicable)
- CKYC Service
- Banking APIs
- Notification Service
- Document Storage
- Audit Logging Service

1.12 Coding Standards

Backend

- Layered Architecture
- RESTful API Design
- DTO Pattern
- Repository Pattern
- Dependency Injection
- Global Exception Handling
- Validation
- Logging (SLF4J)

Frontend

- Component-Based Architecture
- Angular Routing
- Reactive Forms

- HttpClient API Integration
- Route Guards
- Lazy Loading
- Responsive UI using Bootstrap

1.13 REST API Standards

Base URL

/api/v1

Response Format

```
{
  "success": true,
  "message": "Request processed successfully",
  "data": {},
  "timestamp": "2026-07-31T10:30:00"
}
```

HTTP Methods

Method	Purpose
GET	Retrieve Data
POST	Create Record
PUT	Update Record
PATCH	Partial Update
DELETE	Delete Record

1.14 Database Naming Standards

Object	Naming Convention
Tables	snake_case
Columns	snake_case
Primary Key	id
Foreign Key	<table_name>_id
Index	idx_table_column
Unique Key	uk_table_column

1.15 Exception Handling

The application uses centralized exception handling.

Exception	HTTP Status
Bad Request	400
Unauthorized	401
Forbidden	403
Not Found	404
Validation Error	422
Internal Server Error	500

1.16 Deliverables

- Customer Management
- KYC Management
- Product Management
- Quote Management

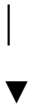
- Underwriting
- Policy Administration
- Premium Management
- Claims Management
- Policy Renewal
- Policy Closure
- Reports & Dashboard
- REST APIs
- Angular UI
- Audit Trail
- Security

1.17 Development Architecture

Angular UI



Controller (REST API)



Service Layer

Repository Layer (JPA)



Hibernate ORM



MySQL Database

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2. Customer Registration Module

Insurance Policy Administration System (PAS)

2.1 Module Overview

The **Customer Registration Module** is the entry point of the Insurance Policy Administration System (PAS). It is responsible for onboarding new customers and maintaining their personal, contact, nominee, address, occupation, and communication details.

This module ensures that every customer receives a **unique Customer ID**, which is used throughout the insurance lifecycle, including **KYC Verification, Quote Generation, Underwriting, Policy Issuance, Premium Payment, and Claim Settlement**.

The module also validates mandatory customer information, prevents duplicate registrations, and stores customer records securely.

2.2 Module Objectives

The Customer Registration Module aims to:

- Register new customers.
- Generate a unique Customer ID.
- Maintain customer profile.
- Store address and contact information.
- Capture nominee details.
- Store occupation and income information.
- Upload customer documents.

- Prevent duplicate customer registrations.
- Enable quick customer search.

2.3 User Roles

Role	Permission
Customer	Self Registration
Agent	Register Customer
Sales Executive	Create Customer
Branch Manager	View Customer
Administrator	Full Access

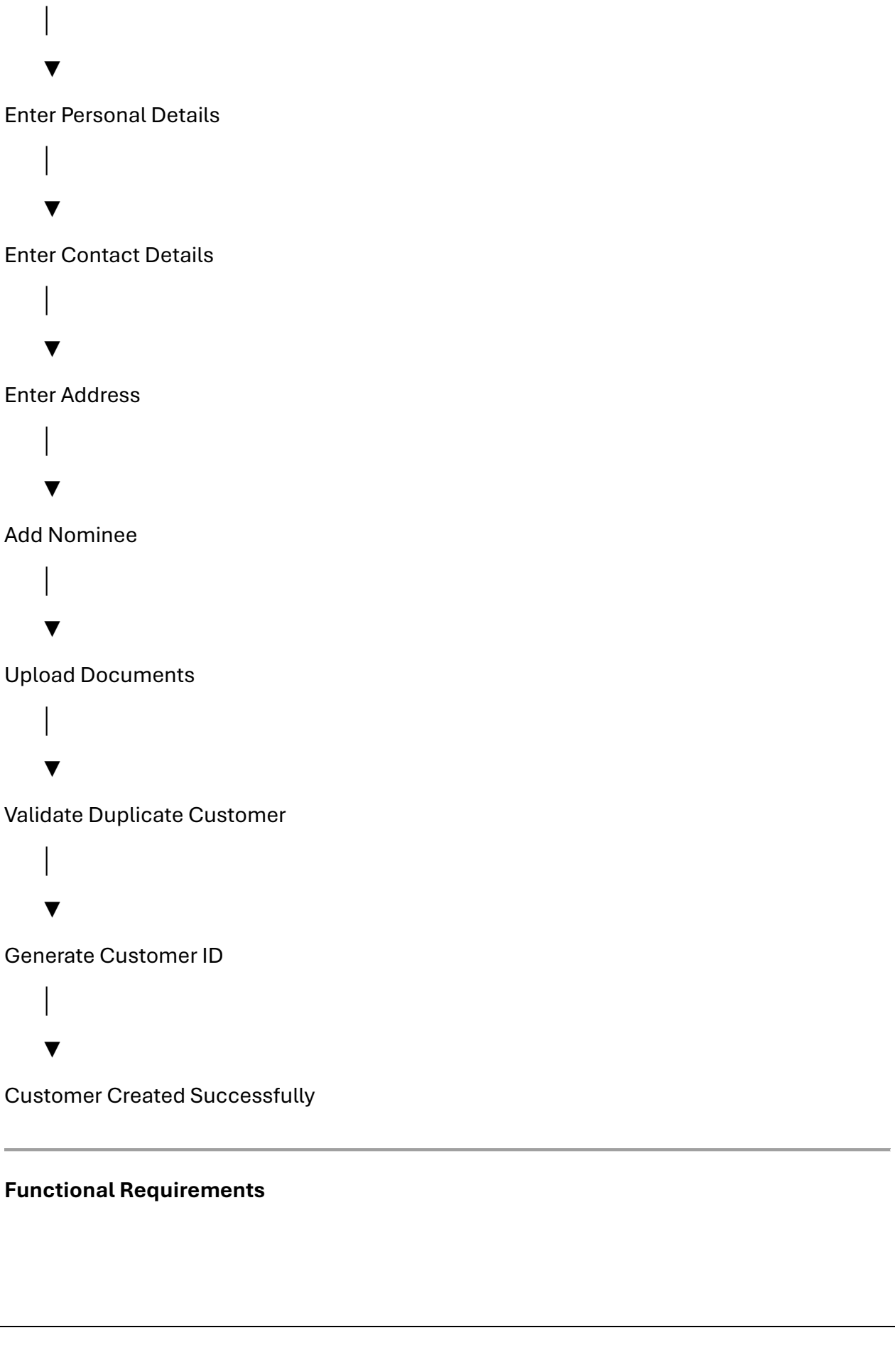
2.4 Module Components

The Customer Registration Module consists of:

1. Customer Registration
2. Personal Details
3. Contact Details
4. Address Details
5. Nominee Details
6. Occupation Details
7. Document Upload
8. Customer Search
9. Customer Profile

2.5 Customer Registration Workflow

Customer Registration



Functional Requirements

FR-CUST-001

Customer Registration

Description

Registers a new customer into the Insurance System.

Customer Information

Field	Type	Mandatory
Customer ID	Auto	Yes
First Name	Text	Yes
Middle Name	Text	No
Last Name	Text	Yes
Date of Birth	Date	Yes
Gender	Dropdown	Yes
Marital Status	Dropdown	Yes
Mobile Number	Text	Yes
Email Address	Email	Yes
Occupation	Dropdown	Yes
Annual Income	Decimal	Yes
Customer Status	Active/Inactive	Yes

Business Rules

- Customer ID is auto-generated.
- Mobile number must be unique.
- Email address must be unique.

- Customer age must comply with product eligibility.
- Mandatory fields cannot be blank.

FR-CUST-002

Contact Information

Description

Stores customer communication details.

Fields

Field	Type
Mobile Number	Text
Alternate Mobile	Text
Email	Email
Office Number	Text

Business Rules

- Mobile number must be verified using OTP.
- Email verification is optional/configurable.
- Contact details can be updated by authorized users.

FR-CUST-003

Address Management

Description

Stores permanent and communication addresses.

Fields

Field	Type
Address Line 1	Text
Address Line 2	Text
City	Dropdown
District	Dropdown
State	Dropdown
Country	Dropdown
Pincode	Number

Business Rules

- Pincode validation is mandatory.
- Permanent and communication addresses may be the same.
- Multiple addresses are supported if required.

FR-CUST-004

Nominee Management

Description

Stores nominee details for policy benefits.

Fields

Field	Type
Nominee Name	Text
Relationship	Dropdown
Date of Birth	Date

Field	Type
Mobile Number	Text
Percentage Share	Decimal

Business Rules

- Total nominee share must equal **100%.**
- At least one nominee is required for eligible products.
- Minor nominees require guardian details.

FR-CUST-005

Occupation Details

Description

Captures occupation information for underwriting and premium calculation.

Fields

Field	Type
Occupation	Dropdown
Employer Name	Text
Employment Type	Dropdown
Annual Income	Decimal
Industry	Dropdown

Business Rules

- Occupation influences risk category.
- Annual income is used during underwriting.

- Employer details are mandatory for salaried applicants.

FR-CUST-006

Document Upload

Description

Uploads customer documents required for KYC and policy processing.

Supported Documents

- Aadhaar Card
- PAN Card
- Passport
- Driving License
- Passport Size Photo
- Signature
- Income Proof
- Address Proof

Business Rules

- Only PDF, JPG, JPEG, and PNG files are allowed.
- Maximum file size is configurable (e.g., 5 MB).
- Uploaded files are stored securely with version tracking.

FR-CUST-007

Customer Search

Description

Searches customers using different criteria.

Search Parameters

- Customer ID
- Mobile Number
- Email Address
- Aadhaar Number
- PAN Number
- Customer Name
- Policy Number

Business Rules

- Partial search is supported.
- Results are filtered based on user role and branch access.

Reports

The Customer Registration Module provides:

- Customer Master Report
- New Customer Registration Report
- Customer Age-wise Report
- City-wise Customer Report
- Occupation-wise Customer Report
- Customer Status Report
- Agent-wise Customer Report

Dashboard

The Customer Dashboard displays:

- Total Customers
- New Customers Today

- Active Customers
- Inactive Customers
- Customers by City
- Customers by Product
- Customers by Agent
- Pending KYC Customers

Database Design

Table INS001 : customer

Column Name	Data Type	Constraint
customer_id	BIGINT	PK
customer_code	VARCHAR(20)	UNIQUE
first_name	VARCHAR(100)	NOT NULL
middle_name	VARCHAR(100)	NULL
last_name	VARCHAR(100)	NOT NULL
date_of_birth	DATE	NOT NULL
gender	ENUM('Male','Female','Other')	NOT NULL
marital_status	ENUM('Single','Married','Divorced','Widowed')	NOT NULL
occupation	VARCHAR(100)	NOT NULL
annual_income	DECIMAL(15,2)	NOT NULL
mobile_number	VARCHAR(15)	UNIQUE
email	VARCHAR(150)	UNIQUE

Column Name	Data Type	Constraint
customer_status	ENUM('Active','Inactive')	DEFAULT 'Active'
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP
updated_at	DATETIME	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Table INS002 : customer_address

Column Name	Data Type	Constraint
address_id	BIGINT	PK
customer_id	BIGINT	FK(Customer)
address_type	ENUM('Permanent','Communication')	
address_line1	VARCHAR(255)	NOT NULL
address_line2	VARCHAR(255)	NULL
city	VARCHAR(100)	NOT NULL
district	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
pincode	VARCHAR(10)	NOT NULL

Table INS003 : nominee

Column Name	Data Type	Constraint
nominee_id	BIGINT	PK
customer_id	BIGINT	FK(Customer)
nominee_name	VARCHAR(150)	NOT NULL
relationship	VARCHAR(50)	NOT NULL
date_of_birth	DATE	NOT NULL
mobile_number	VARCHAR(15)	NULL
share_percentage	DECIMAL(5,2)	NOT NULL

Table INS004 : customer_document

Column Name	Data Type	Constraint
document_id	BIGINT	PK
customer_id	BIGINT	FK(Customer)
document_type	VARCHAR(100)	NOT NULL
file_name	VARCHAR(255)	NOT NULL
file_path	VARCHAR(500)	NOT NULL
uploaded_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
verification_status	ENUM('Pending','Verified','Rejected')	DEFAULT 'Pending'

Table INS005 : customer_contact

Column Name	Data Type	Constraint
contact_id	BIGINT	PK

Column Name	Data Type	Constraint
customer_id	BIGINT	FK(Customer)
alternate_mobile	VARCHAR(15)	NULL
office_phone	VARCHAR(20)	NULL
preferred_contact_mode	ENUM('Mobile','Email','SMS')	DEFAULT 'Mobile'

RESTful API Endpoints

Customer APIs

Method	Endpoint	Description
POST	/api/v1/customers	Register New Customer
GET	/api/v1/customers	Get All Customers
GET	/api/v1/customers/{id}	Get Customer By ID
PUT	/api/v1/customers/{id}	Update Customer
DELETE	/api/v1/customers/{id}	Delete Customer
GET	/api/v1/customers/search	Search Customer

Address APIs

Method	Endpoint	Description
POST	/api/v1/customers/{id}/addresses	Add Address
GET	/api/v1/customers/{id}/addresses	List Addresses
PUT	/api/v1/addresses/{addressId}	Update Address
DELETE	/api/v1/addresses/{addressId}	Delete Address

Nominee APIs

Method	Endpoint	Description
POST	/api/v1/customers/{id}/nominees	Add Nominee
GET	/api/v1/customers/{id}/nominees	List Nominees
PUT	/api/v1/nominees/{nomineeId}	Update Nominee
DELETE	/api/v1/nominees/{nomineeId}	Delete Nominee

Document APIs

Method	Endpoint	Description
POST	/api/v1/customers/{id}/documents	Upload Document
GET	/api/v1/customers/{id}/documents	List Documents
DELETE	/api/v1/documents/{documentId}	Delete Document
PUT	/api/v1/documents/{documentId}/verify	Verify Document

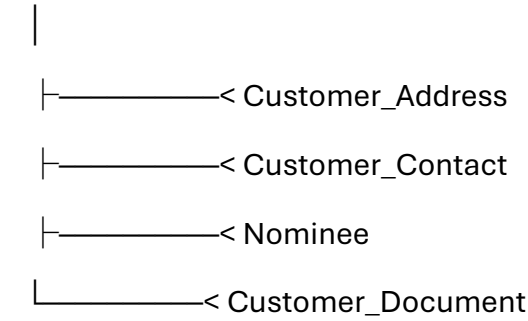
Angular Screens

- Customer Registration

- Customer List
- Customer Details
- Customer Search
- Address Management
- Nominee Management
- Document Upload
- Customer Profile

Entity Relationship

Customer (1)



Security Features

- JWT-based authentication.
- Role-Based Access Control (RBAC).
- Server-side validation.
- Duplicate customer detection.
- Secure document storage.
- Audit trail for customer updates.
- Input validation and SQL injection protection.

Business Rules

- Customer Code is auto-generated and unique.
- Mobile number and Email ID must be unique.
- Nominee share percentage must total **100%**.
- Mandatory documents are verified before KYC approval.
- Customer information is editable only by authorized roles.
- All create/update/delete operations are recorded in the audit log.

Deliverables

- Customer Registration
- Customer Profile Management
- Contact & Address Management
- Nominee Management
- Document Upload
- Customer Search
- Reports & Dashboard
- REST APIs
- Angular UI
- Audit Trail

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3. KYC Verification Module

Insurance Policy Administration System (PAS)

3.1 Module Overview

The **Know Your Customer (KYC) Verification Module** is a mandatory compliance module in the Insurance Policy Administration System (PAS). It verifies the customer's identity, address, age, and financial information before allowing policy issuance.

The KYC process ensures compliance with **IRDAI (Insurance Regulatory and Development Authority of India)** guidelines and Anti-Money Laundering (AML) regulations. The module supports document verification, CKYC integration, document approval/rejection, and audit logging.

A policy cannot proceed to **Quote Generation** or **Underwriting** until KYC is successfully verified.

3.2 Module Objectives

The KYC Verification Module aims to:

- Verify customer identity.
- Validate government-issued documents.
- Prevent duplicate KYC.
- Integrate with CKYC services.
- Store verified KYC records.
- Approve or reject submitted documents.
- Maintain complete audit history.
- Ensure regulatory compliance.

3.3 User Roles

Role	Permission
Customer	Upload KYC Documents
Agent	Submit KYC
KYC Executive	Verify Documents
Compliance Officer	Approve/Reject KYC
Branch Manager	View KYC Status

Role	Permission
Administrator	Configure KYC Rules

3.4 Module Components

The KYC Verification Module consists of:

1. KYC Registration
2. Aadhaar Verification
3. PAN Verification
4. Passport Verification
5. Driving License Verification
6. Address Proof Verification
7. CKYC Integration
8. KYC Approval/Rejection
9. KYC Audit Log

3.5 KYC Verification Workflow

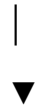
Customer Registration



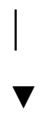
Upload KYC Documents



Document Validation



CKYC / Identity Verification



KYC Executive Review



Approve / Reject



KYC Status Updated



Proceed to Product Selection

Functional Requirements

FR-KYC-001

KYC Registration

Description

Creates a KYC request linked to the customer profile.

KYC Information

Field	**Type**	**Mandatory**
Customer ID	Auto	Yes
KYC Number	Auto	Yes
KYC Date	Date	Yes

Field	Type	Mandatory
Verification Status	Dropdown	Yes
Remarks	Text	No

Business Rules

- One active KYC record per customer.
- KYC Number is auto-generated.
- KYC request is created after customer registration.

FR-KYC-002

Aadhaar Verification

Description

Verifies Aadhaar details submitted by the customer.

Fields

Field	Type
Aadhaar Number	Text
Name	Text
Date of Birth	Date
Address	Text

Business Rules

- Aadhaar must contain 12 digits.
- Duplicate Aadhaar numbers are not allowed.
- Aadhaar verification status must be recorded.

FR-KYC-003

PAN Verification

Description

Validates PAN details for identity and tax compliance.

Fields

Field	Type
PAN Number	Text
PAN Holder Name	Text
PAN Status	Verified/Rejected

Business Rules

- PAN format must follow Indian standards (e.g., ABCDE1234F).
- PAN must be unique per customer.

FR-KYC-004

Passport / Driving License Verification

Description

Supports verification of alternate identity documents.

Supported Documents

- Passport
- Driving License
- Voter ID
- Government Identity Card

Business Rules

- Expired documents are rejected.
- Document number must be unique.

FR-KYC-005

Address Verification

Description

Verifies the customer's residential address.

Accepted Address Proofs

- Aadhaar Card
- Passport
- Electricity Bill
- Water Bill
- Bank Statement
- Gas Connection Bill

Business Rules

- Address proof should not be older than the configured validity period (e.g., 3 months for utility bills).
- Address must match customer profile or be updated after verification.

FR-KYC-006

CKYC Integration

Description

Integrates with the Central KYC (CKYC) registry to validate customer records.

Fields

Field	Type
CKYC Number	Text
CKYC Status	Verified/Not Found
Verification Date	Date

Business Rules

- CKYC lookup is optional/configurable.
- If CKYC exists, customer data can be auto-populated.

FR-KYC-007

KYC Approval / Rejection

Description

Approves or rejects KYC requests after document verification.

Approval Status

- Pending
- Verified
- Rejected
- Re-Submitted

Business Rules

- Only KYC Executive or Compliance Officer can approve/reject.
- Rejected KYC must include rejection reason.
- Customer can re-upload rejected documents.

FR-KYC-008

Audit Log

Description

Maintains complete history of KYC actions.

Logged Activities

- Document Upload
- Verification
- Approval
- Rejection
- Re-Submission
- Status Change

Business Rules

- Audit records cannot be modified or deleted.
- Every action records user, timestamp, and remarks.

Reports

The KYC Verification Module provides:

- Pending KYC Report
- Verified KYC Report
- Rejected KYC Report
- CKYC Verification Report
- Branch-wise KYC Report
- Agent-wise KYC Report
- KYC Turnaround Time Report

Dashboard

The KYC Dashboard displays:

- Total KYC Requests
- Pending Verification
- Verified KYC
- Rejected KYC
- Re-Submitted KYC
- CKYC Success Rate
- Daily Verification Count

Database Design

Table INS006 : customer_kyc

Column Name	Data Type	Constraint
kyc_id	BIGINT	PK
customer_id	BIGINT	FK(Customer)
kyc_number	VARCHAR(30)	UNIQUE
kyc_date	DATE	NOT NULL
verification_status	ENUM('Pending','Verified','Rejected','Re-Submitted')	DEFAULT 'Pending'
remarks	VARCHAR(500)	NULL
verified_by	BIGINT	FK(Employee)
verified_date	DATETIME	NULL

Table INS007 : kyc_document

Column Name	Data Type	Constraint
document_id	BIGINT	PK
kyc_id	BIGINT	FK(Customer_KYC)
document_type	ENUM('Aadhaar','PAN','Passport','Driving License','Voter ID','Address Proof')	
document_number	VARCHAR(100)	NOT NULL
file_path	VARCHAR(500)	NOT NULL
upload_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
verification_status	ENUM('Pending','Verified','Rejected')	

Table INS008 : ckyc_details

Column Name	Data Type	Constraint
ckyc_id	BIGINT	PK
customer_id	BIGINT	FK(Customer)
ckyc_number	VARCHAR(30)	UNIQUE
verification_date	DATETIME	
ckyc_status	ENUM('Verified','Not Found','Pending')	

Table INS009 : kyc_verification_log

Column Name	Data Type	Constraint
log_id	BIGINT	PK
kyc_id	BIGINT	FK(Customer_KYC)

Column Name	Data Type	Constraint
action	VARCHAR(100)	
performed_by	BIGINT	FK(Employee)
action_date	DATETIME	DEFAULT CURRENT_TIMESTAMP
remarks	VARCHAR(500)	

Table INS010 : document_type_master

Column Name	Data Type	Constraint
document_type_id	BIGINT	PK
document_name	VARCHAR(100)	UNIQUE
mandatory_flag	BOOLEAN	
status	ENUM('Active','Inactive')	

RESTful API Endpoints

KYC APIs

Method	Endpoint	Description
POST	/api/v1/kyc	Create KYC Request
GET	/api/v1/kyc	List KYC Requests
GET	/api/v1/kyc/{kycId}	Get KYC Details
PUT	/api/v1/kyc/{kycId}	Update KYC
DELETE	/api/v1/kyc/{kycId}	Delete KYC Request

Document Verification APIs

Method	Endpoint	Description
POST	/api/v1/kyc/documents	Upload KYC Document
GET	/api/v1/kyc/documents/{documentId}	View Document
PUT	/api/v1/kyc/documents/{documentId}/verify	Verify Document
PUT	/api/v1/kyc/documents/{documentId}/reject	Reject Document

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CKYC APIs

Method	Endpoint	Description
POST	/api/v1/kyc/ckyc/verify	Verify CKYC
GET	/api/v1/kyc/ckyc/{customerId}	Get CKYC Details

Approval APIs

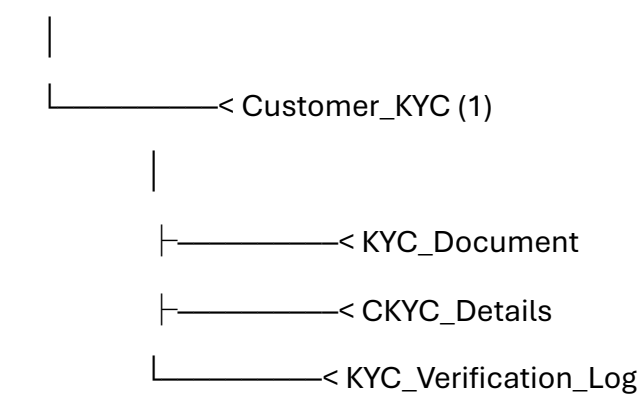
Method	Endpoint	Description
PUT	/api/v1/kyc/{kycId}/approve	Approve KYC
PUT	/api/v1/kyc/{kycId}/reject	Reject KYC
PUT	/api/v1/kyc/{kycId}/resubmit	Re-submit KYC

Angular Screens

- KYC Dashboard
- KYC Request List
- KYC Verification Screen
- Document Upload
- Document Viewer

- CKYC Verification
- KYC Approval/Rejection
- KYC Audit Log

Entity Relationship



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Secure file upload with virus/type validation
- Encrypted storage of sensitive document references
- Audit logging for every KYC action
- Input validation to prevent malicious data

Business Rules

- Customer registration must be completed before KYC initiation.
- A customer cannot have more than one active KYC request.
- Quote generation and underwriting are blocked until KYC is verified.
- Rejected documents can be re-uploaded and re-submitted.

- Every approval, rejection, or modification is recorded in the audit log.
- Only authorized users can approve or reject KYC.

Deliverables

- KYC Registration
- Aadhaar/PAN/Passport Verification
- Address Proof Verification
- CKYC Integration
- KYC Approval Workflow
- Audit Log
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Security & Compliance

---

4. Insurance Product Management Module

Insurance Policy Administration System (PAS)

4.1 Module Overview

The **Insurance Product Management Module** is the master module of the Insurance Policy Administration System (PAS). It enables administrators and product managers to create, configure, and manage insurance products offered by the organization.

Each insurance product contains predefined business rules such as **minimum and maximum age**, **policy term**, **premium frequency**, **coverage amount**, **riders**, **waiting period**, **claim rules**, **grace period**, **GST**, and **commission structure**.

This module acts as the foundation for **Quote Generation**, **Premium Calculation**, **Underwriting**, **Policy Issuance**, **Renewal**, and **Claims Management**.

4.2 Module Objectives

The Insurance Product Management Module aims to:

- Create insurance products.
- Configure product categories.
- Define eligibility criteria.
- Configure premium frequencies.
- Configure sum insured limits.
- Configure policy terms.
- Manage riders and benefits.
- Configure waiting periods.
- Maintain commission structure.
- Activate or deactivate insurance products.

4.3 User Roles

Role	Permission
Product Manager	Create & Configure Products
Administrator	Full Access
Underwriter	View Product Rules
Agent	View Products
Customer	View Active Products

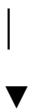
4.4 Module Components

The Product Management Module consists of:

1. Product Master
2. Product Category
3. Coverage Management
4. Benefit Management
5. Rider Management
6. Sum Insured Configuration
7. Eligibility Rules
8. Premium Frequency
9. Commission Configuration
10. Product Activation

4.5 Product Management Workflow

Administrator



Create Product Category



Create Insurance Product



Configure Coverage



Configure Benefits



Functional Requirements

FR-PROD-001

Product Master

Description

Creates and maintains insurance products.

Product Information

Field	Type	Mandatory
Product ID	Auto	Yes

Field	Type	Mandatory
Product Code	Auto	Yes
Product Name	Text	Yes
Product Category	Dropdown	Yes
Product Type	Dropdown	Yes
Description	TextArea	No
Status	Active/Inactive	Yes

Business Rules

- Product Code must be unique.
- Product Name cannot be duplicated.
- Inactive products cannot be sold.
- Product deletion is not allowed if policies already exist.

FR-PROD-002

Product Category

Description

Groups insurance products into categories.

Categories

- Life Insurance
- Health Insurance
- Motor Insurance
- Travel Insurance
- Property Insurance
- Home Insurance

- Marine Insurance
- Personal Accident Insurance
- Term Insurance
- Child Plan

Business Rules

- Category must exist before product creation.
- One category can contain multiple products.

FR-PROD-003

Coverage Management

Description

Defines the insurance cover available under each product.

Coverage Fields

Field	**Type**
Coverage Name	Text
Coverage Amount	Decimal
Minimum Coverage	Decimal
Maximum Coverage	Decimal
Waiting Period	Number
Deductible	Decimal

Business Rules

- Coverage amount must fall within configured limits.
- Waiting period is product-specific.

FR-PROD-004

Benefit Management

Description

Defines policy benefits.

Benefits

- Hospital Cash
- Critical Illness
- Accidental Death
- Disability Benefit
- Maturity Benefit
- Survival Benefit
- Family Benefit

Business Rules

- Benefits vary by product.
- Benefits are displayed during quote generation.

FR-PROD-005

Rider Management

Description

Configures optional riders.

Riders

- Accidental Rider
- Critical Illness Rider
- Waiver of Premium Rider

- Disability Rider
- Income Benefit Rider

Business Rules

- Riders are optional.
- Multiple riders can be attached.
- Rider premium is calculated separately.

FR-PROD-006

Eligibility Rules

Description

Defines customer eligibility.

Rules

Field	Value
Minimum Age	Configurable
Maximum Age	Configurable
Minimum Income	Configurable
Occupation Category	Configurable
Medical Required	Yes/No

Business Rules

- Customer age must satisfy eligibility.
- Medical examination may be mandatory.

FR-PROD-007

Sum Insured Configuration

Description

Defines minimum and maximum coverage.

Example

Product	Minimum	Maximum
Health	₹50,000	₹50,00,000
Motor	₹1,00,000	₹2,00,00,000

Business Rules

- Quote cannot exceed maximum limit.
- Minimum sum insured must be validated.

FR-PROD-008

Premium Frequency

Description

Defines premium payment frequency.

Supported Frequencies

- Monthly
- Quarterly
- Half-Yearly
- Yearly
- Single Premium

Business Rules

- Frequency depends on product.

- Premium schedule is generated accordingly.

FR-PROD-009

Commission Configuration

Description

Defines commission payable to agents.

Fields

Field	Type
Agent Commission	Percentage
Branch Commission	Percentage
Renewal Commission	Percentage

Business Rules

- Commission depends on product category.
- Renewal commission may differ from first-year commission.

Reports

The Product Management Module provides:

- Product Master Report
- Category-wise Product Report
- Coverage Report
- Benefit Report
- Rider Report
- Active Product Report
- Inactive Product Report

- Commission Report

Dashboard

The Product Dashboard displays:

- Total Products
- Active Products
- Inactive Products
- Products by Category
- Products with Riders
- Products Awaiting Approval

Database Design

Table INS011 : product_category

Column Name	Data Type	Constraint
category_id	BIGINT	PK
category_name	VARCHAR(100)	UNIQUE
description	VARCHAR(255)	NULL
status	ENUM('Active','Inactive')	

Table INS012 : insurance_product

Column Name	Data Type	Constraint
product_id	BIGINT	PK
category_id	BIGINT	FK(Product_Category)

Column Name	Data Type	Constraint
product_code	VARCHAR(30)	UNIQUE
product_name	VARCHAR(150)	NOT NULL
product_type	VARCHAR(100)	NOT NULL
description	TEXT	
minimum_age	INT	
maximum_age	INT	
minimum_sum_insured	DECIMAL(15,2)	
maximum_sum_insured	DECIMAL(15,2)	
status	ENUM('Active','Inactive')	

Table INS013 : coverage_master

Column Name	Data Type	Constraint
coverage_id	BIGINT	PK
product_id	BIGINT	FK(Insurance_Product)
coverage_name	VARCHAR(150)	
minimum_coverage	DECIMAL(15,2)	
maximum_coverage	DECIMAL(15,2)	
waiting_period_days	INT	
deductible_amount	DECIMAL(15,2)	

Table INS014 : rider_master

Column Name	Data Type	Constraint
rider_id	BIGINT	PK
product_id	BIGINT	FK(Insurance_Product)
rider_name	VARCHAR(150)	
rider_description	TEXT	
rider_premium	DECIMAL(15,2)	
status	ENUM('Active','Inactive')	

Table INS015 : benefit_master

Column Name	Data Type	Constraint
benefit_id	BIGINT	PK
product_id	BIGINT	FK(Insurance_Product)
benefit_name	VARCHAR(150)	
benefit_description	TEXT	
benefit_limit	DECIMAL(15,2)	

Table INS016 : premium_frequency

Column Name	Data Type	Constraint
frequency_id	BIGINT	PK
frequency_name	ENUM('Monthly','Quarterly','Half-Yearly','Yearly','Single Premium')	
installment_count	INT	
status	ENUM('Active','Inactive')	

Table INS017 : product_commission

Column Name	Data Type	Constraint
commission_id	BIGINT	PK
product_id	BIGINT	FK(Insurance_Product)
first_year_commission	DECIMAL(5,2)	
renewal_commission	DECIMAL(5,2)	
branch_commission	DECIMAL(5,2)	

RESTful API Endpoints

Product APIs

Method	Endpoint	Description
POST	/api/v1/products	Create Product
GET	/api/v1/products	Get All Products
GET	/api/v1/products/{id}	Get Product By ID
PUT	/api/v1/products/{id}	Update Product
DELETE	/api/v1/products/{id}	Deactivate Product

Category APIs

Method	Endpoint	Description
POST	/api/v1/product-categories	Create Category
GET	/api/v1/product-categories	List Categories

Method	Endpoint	Description
PUT	/api/v1/product-categories/{id}	Update Category
DELETE	/api/v1/product-categories/{id}	Deactivate Category

Coverage APIs

Method	Endpoint	Description
POST	/api/v1/products/{id}/coverages	Add Coverage
GET	/api/v1/products/{id}/coverages	List Coverages
PUT	/api/v1/coverages/{coverageId}	Update Coverage
DELETE	/api/v1/coverages/{coverageId}	Delete Coverage

Rider APIs

Method	Endpoint	Description
POST	/api/v1/products/{id}/riders	Add Rider
GET	/api/v1/products/{id}/riders	List Riders
PUT	/api/v1/riders/{riderId}	Update Rider
DELETE	/api/v1/riders/{riderId}	Delete Rider

Benefit APIs

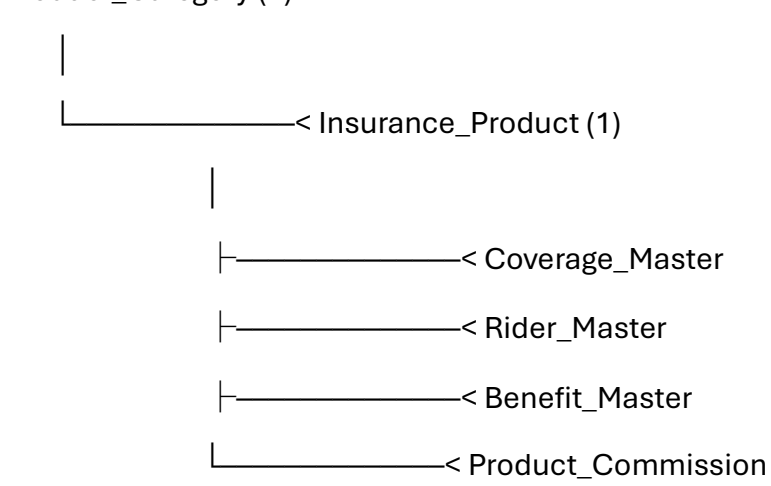
Method	Endpoint	Description
POST	/api/v1/products/{id}/benefits	Add Benefit
GET	/api/v1/products/{id}/benefits	List Benefits

Method	Endpoint	Description
PUT	/api/v1/benefits/{benefitId}	Update Benefit
DELETE	/api/v1/benefits/{benefitId}	Delete Benefit

Angular Screens

- Product Dashboard
- Product Master
- Product Category
- Coverage Master
- Rider Master
- Benefit Master
- Premium Frequency Master
- Commission Configuration
- Product Details

Entity Relationship



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Product versioning for changes
- Audit log for all product configuration updates
- Validation of eligibility and coverage limits
- Secure API access using Spring Security

Business Rules

- Product code and category name must be unique.
- Only active products are available for quote generation.
- Eligibility rules are validated before quote creation.
- Sum insured must fall within configured limits.
- Rider premium is added separately to the base premium.
- Product deletion is not allowed if policies have been issued.
- Commission rules are product-specific and configurable.
- All product changes are captured in the audit trail.

Deliverables

- Product Category Management
- Insurance Product Master
- Coverage Management
- Benefit Management
- Rider Management
- Sum Insured Configuration
- Eligibility Rule Management

- Premium Frequency Configuration
- Commission Configuration
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

---

5. Quote Generation & Premium Calculation Module

Insurance Policy Administration System (PAS)

5.1 Module Overview

The **Quote Generation & Premium Calculation Module** is one of the most critical modules in the Insurance Policy Administration System (PAS). It calculates the insurance premium based on customer information, selected insurance product, sum insured, policy term, age, occupation, risk profile, riders, discounts, taxes, and underwriting rules.

The generated quote is valid for a configurable period (for example, **30 days**) and can be converted into an insurance policy after underwriting approval.

This module is integrated with:

- Customer Registration
- KYC Verification
- Product Management
- Underwriting
- Policy Issuance
- Premium Schedule

5.2 Module Objectives

The Quote Generation Module aims to:

- Generate insurance quotations.
- Calculate premium automatically.
- Calculate taxes (GST).
- Apply discounts.
- Apply risk loading.
- Include optional riders.
- Generate quote number.
- Maintain quote history.
- Convert approved quote into policy.

5.3 User Roles

Role	**Permission**
Customer	Request Quote
Agent	Generate Quote
Sales Executive	Create/Edit Quote
Underwriter	Review Quote
Branch Manager	Approve High Value Quote
Administrator	Configure Premium Rules

5.4 Module Components

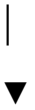
The Quote Generation Module consists of:

1. Quote Generation
2. Premium Calculation Engine
3. Risk Assessment
4. Discount Management

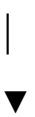
5. Rider Premium
6. GST Calculation
7. Quote Approval
8. Quote Expiry
9. Quote Conversion

5.5 Quote Generation Workflow

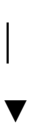
Customer Registration



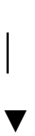
KYC Verified



Select Insurance Product



Enter Sum Insured



Select Policy Term



Select Riders



Risk Assessment



Functional Requirements

FR-QT-001

Quote Generation

Description

Generates an insurance quotation based on customer and product details.

Quote Information

Field	Type	Mandatory
Quote Number	Auto	Yes
Customer ID	FK	Yes
Product	Dropdown	Yes
Policy Term	Number	Yes
Sum Insured	Decimal	Yes
Premium Frequency	Dropdown	Yes
Quote Date	Date	Yes
Quote Expiry Date	Date	Yes
Quote Status	Pending/Approved/Rejected	Yes

Business Rules

- Quote Number is auto-generated.
- Quote validity is configurable (default 30 days).
- Expired quotes cannot be converted into policies.

FR-QT-002

Premium Calculation Engine

Description

Automatically calculates premium using configured business rules.

Premium Formula

Base Premium

+

Risk Loading

+

Rider Premium

-

Discount

+

GST

=

Final Premium

Example Formula

Base Premium = (Sum Insured × Product Rate) ÷ 1000

Risk Loading = Base Premium × Risk %

Rider Premium = Sum of Selected Riders

Discount = Base Premium × Discount %

GST = (Premium After Discount) × GST %

Final Premium = Base Premium

+ Risk Loading

+ Rider Premium

- Discount

+ GST

Example Calculation

Item	Amount
Sum Insured	₹10,00,000
Product Rate	₹2.50
Base Premium	₹2,500
Risk Loading (10%)	₹250
Rider Premium	₹600
Discount	₹200
GST (18%)	₹567
Final Premium	₹3,717

FR-QT-003

Risk Assessment

Description

Calculates customer risk before premium calculation.

Risk Parameters

Parameter	Impact
Age	High
Gender	Medium
Occupation	High
Smoking	High

Parameter	Impact
Alcohol	Medium
Medical History	High
Existing Diseases	High
Previous Claims	Medium

Risk Categories

Score	Risk
0–30	Low
31–70	Medium
71–100	High

Business Rules

- High-risk customers require underwriting approval.
- Low-risk quotes are generated immediately.

FR-QT-004

Discount Management

Description

Applies discounts based on company policy.

Supported Discounts

- Online Purchase Discount
- Corporate Discount

- Employee Discount
- No Claim Bonus (NCB)
- Festival Offer
- Loyalty Discount

Business Rules

- Multiple discounts may be combined if allowed.
- Maximum discount percentage is configurable.

FR-QT-005

Rider Premium

Description

Calculates premium for optional riders.

Sample Riders

Rider	Premium
Critical Illness	₹500
Accident Rider	₹350
Disability Rider	₹450
Waiver of Premium	₹300

Business Rules

- Rider premium is added separately.
- Rider eligibility depends on the selected product.

FR-QT-006

GST Calculation

Description

Calculates GST on the premium.

Formula

GST = Taxable Premium × GST Rate

Business Rules

- GST rate is configurable.
- GST is applied after discounts.

FR-QT-007

Quote Approval

Description

Approves or rejects generated quotes.

Status

- Pending
- Approved
- Rejected
- Expired
- Converted

Business Rules

- High-value quotes require manager approval.

- Underwriting approval is mandatory for high-risk customers.

FR-QT-008

Quote Conversion

Description

Converts approved quotes into insurance policies.

Business Rules

- Only approved quotes can be converted.
- Expired quotes cannot be converted.
- Customer KYC must be verified.
- Premium payment must be initiated before policy issuance.

Reports

The Quote Generation Module provides:

- Daily Quote Report
- Product-wise Quote Report
- Agent-wise Quote Report
- Quote Conversion Report
- Expired Quote Report
- Premium Collection Forecast
- Discount Utilization Report

Dashboard

The Quote Dashboard displays:

- Total Quotes

- Approved Quotes
- Pending Quotes
- Rejected Quotes
- Expired Quotes
- Converted Policies
- Total Premium Value
- Average Premium Amount

Database Design

Table INS018 : quote

Column Name	Data Type	Constraint
quote_id	BIGINT	PK
quote_number	VARCHAR(30)	UNIQUE
customer_id	BIGINT	FK(Customer)
product_id	BIGINT	FK(Insurance_Product)
policy_term	INT	NOT NULL
sum_insured	DECIMAL(15,2)	NOT NULL
premium_frequency_id	BIGINT	FK(Premium_Frequency)
quote_date	DATE	NOT NULL
expiry_date	DATE	NOT NULL
quote_status	ENUM('Pending','Approved','Rejected','Expired','Converted')	

Table INS019 : premium_calculation

Column Name	Data Type	Constraint
calculation_id	BIGINT	PK
quote_id	BIGINT	FK(Quote)
base_premium	DECIMAL(15,2)	
risk_loading	DECIMAL(15,2)	
rider_premium	DECIMAL(15,2)	
discount_amount	DECIMAL(15,2)	
gst_amount	DECIMAL(15,2)	
final_premium	DECIMAL(15,2)	

Table INS020 : risk_assessment

Column Name	Data Type	Constraint
risk_id	BIGINT	PK
quote_id	BIGINT	FK(Quote)
age_score	INT	
occupation_score	INT	
medical_score	INT	
lifestyle_score	INT	
total_score	INT	
risk_category	ENUM('Low','Medium','High')	

Table INS021 : quote_rider

Column Name	Data Type	Constraint
rider_selection_id	BIGINT	PK
quote_id	BIGINT	FK(Quote)
rider_id	BIGINT	FK(Rider_Master)
rider_premium	DECIMAL(15,2)	

Table INS022 : discount_master

Column Name	Data Type	Constraint
discount_id	BIGINT	PK
discount_name	VARCHAR(100)	
discount_percentage	DECIMAL(5,2)	
valid_from	DATE	
valid_to	DATE	
status	ENUM('Active','Inactive')	

Table INS023 : quote_history

Column Name	Data Type	Constraint
history_id	BIGINT	PK
quote_id	BIGINT	FK(Quote)
action	VARCHAR(100)	
performed_by	BIGINT	FK(Employee)

Column Name	Data Type	Constraint
action_date	DATETIME	
remarks	VARCHAR(500)	

RESTful API Endpoints

Quote APIs

Method	Endpoint	Description
POST	/api/v1/quotes	Generate Quote
GET	/api/v1/quotes	Get All Quotes
GET	/api/v1/quotes/{id}	Get Quote By ID
PUT	/api/v1/quotes/{id}	Update Quote
DELETE	/api/v1/quotes/{id}	Cancel Quote

Premium Calculation APIs

Method	Endpoint	Description
POST	/api/v1/quotes/calculate-premium	Calculate Premium
GET	/api/v1/quotes/{id}/premium	Get Premium Details

Risk Assessment APIs

Method	Endpoint	Description
POST	/api/v1/risk-assessment	Assess Customer Risk
GET	/api/v1/risk-assessment/{quoteId}	Get Risk Assessment

Quote Approval APIs

Method	Endpoint	Description
PUT	/api/v1/quotes/{id}/approve	Approve Quote
PUT	/api/v1/quotes/{id}/reject	Reject Quote
PUT	/api/v1/quotes/{id}/convert	Convert Quote to Policy

Angular Screens

- Quote Dashboard
- New Quote
- Quote List
- Quote Details
- Premium Calculator
- Rider Selection
- Risk Assessment
- Quote Approval
- Quote Conversion

Entity Relationship

Customer (1)

```
    │
    └──────────< Quote (1)
                    │
                    ├──────────< Premium_Calculation
                    ├──────────< Risk_Assessment
                    ├──────────< Quote_Rider
```

└──────────< Quote_History

Insurance_Product (1)

│

└──────────< Quote

Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Premium calculation rules secured on the server side
- Audit trail for all quote modifications
- Validation of discounts and rider eligibility
- Secure REST APIs with Spring Security

Business Rules

- KYC verification must be completed before quote generation.
- Only active insurance products can be quoted.
- Sum insured must fall within the configured product limits.
- High-risk quotes require underwriting approval.
- Quote validity is configurable (e.g., 30 days).
- Expired quotes cannot be converted into policies.
- GST is calculated after applying eligible discounts.
- Every quote calculation and status change is recorded in the audit log.

Deliverables

- Quote Generation

- Premium Calculation Engine
- Risk Assessment
- Discount Management
- Rider Premium Calculation
- GST Calculation
- Quote Approval Workflow
- Quote-to-Policy Conversion
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

---

6. Underwriting Module

Insurance Policy Administration System (PAS)

6.1 Module Overview

The **Underwriting Module** is the decision-making engine of the Insurance Policy Administration System (PAS). It evaluates the customer's risk profile before issuing an insurance policy.

The underwriting process determines whether the insurance company should:

- Accept the proposal
- Accept with additional premium (Loading)
- Accept with exclusions
- Request additional documents
- Request medical examination
- Reject the proposal

This module evaluates customer information collected during **Customer Registration**, **KYC Verification**, and **Quote Generation** to make a final underwriting decision.

The underwriting process may be **Automatic (Rule-Based)** or **Manual (Underwriter Decision)** depending on the product, risk score, and underwriting guidelines.

6.2 Module Objectives

The Underwriting Module aims to:

- Assess customer risk.
- Evaluate medical and financial information.
- Detect high-risk customers.
- Apply underwriting rules.
- Recommend policy acceptance or rejection.
- Calculate additional premium (Loading).
- Manage underwriting decisions.
- Maintain complete audit logs.

6.3 User Roles

Role	**Permission**
Underwriter	Review & Decide Cases
Senior Underwriter	Approve High-Risk Cases
Medical Officer	Review Medical Reports
Branch Manager	Approve Special Cases
Agent	View Underwriting Status
Customer	View Application Status
Administrator	Configure Underwriting Rules

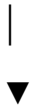
6.4 Module Components

The Underwriting Module consists of:

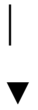
1. Underwriting Case Creation
2. Medical Underwriting
3. Financial Underwriting
4. Occupational Risk Assessment
5. Lifestyle Assessment
6. Risk Scoring Engine
7. Loading Calculation
8. Underwriting Decision
9. Audit Log

6.5 Underwriting Workflow

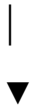
Quote Generated



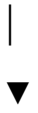
Create Underwriting Case



Medical Evaluation



Financial Evaluation



Occupation Risk Assessment



Lifestyle Risk Assessment



Calculate Risk Score



Automatic Rule Engine



Manual Underwriter Review (if required)



Approve / Reject / Additional Loading



Proceed to Policy Issuance

Functional Requirements

FR-UW-001

Underwriting Case

Description

Creates an underwriting case after successful quote generation.

Fields

Field	Type	Mandatory
Underwriting ID	Auto	Yes
Quote Number	FK	Yes
Customer ID	FK	Yes
Product ID	FK	Yes
Underwriting Date	Date	Yes
Status	Pending/Approved/Rejected	Yes

Business Rules

- One underwriting case per quote.
- Case ID is auto-generated.
- Only approved quotes proceed to underwriting.

FR-UW-002

Medical Underwriting

Description

Evaluates the customer's health profile.

Medical Details

Field	Type
Height	Decimal
Weight	Decimal
BMI	Decimal

Field	Type
Blood Pressure	Text
Diabetes	Yes/No
Heart Disease	Yes/No
Cancer History	Yes/No
Smoking	Yes/No
Alcohol Consumption	Yes/No
Medical Report	Upload

Business Rules

- BMI is calculated automatically.
- Medical reports are mandatory for high-value policies.
- High-risk diseases increase premium loading.

FR-UW-003

Financial Underwriting

Description

Evaluates customer's financial capability.

Financial Details

Field	Type
Annual Income	Decimal
Monthly Income	Decimal
Occupation	Text

Field	Type
Employer	Text
Existing Insurance	Decimal
Loan Liability	Decimal

Business Rules

- Sum insured should not exceed the customer's financial eligibility.
- Financial documents are mandatory above configurable limits.

FR-UW-004

Occupation Risk Assessment

Description

Evaluates occupational hazards.

Risk Categories

Occupation	Risk
Software Engineer	Low
Teacher	Low
Doctor	Medium
Police Officer	High
Firefighter	High
Miner	Very High
Pilot	High

Business Rules

- Risk category affects premium loading.
- Hazardous occupations may require manual approval.

FR-UW-005

Lifestyle Assessment

Description

Evaluates personal habits affecting insurance risk.

Parameters

- Smoking
- Alcohol Consumption
- Adventure Sports
- Frequent International Travel
- Dangerous Hobbies

Business Rules

- Smoking increases premium.
- Adventure sports may require exclusions.
- Lifestyle information contributes to risk score.

FR-UW-006

Risk Scoring Engine

Description

Calculates the overall underwriting risk score.

Sample Scoring

Parameter	Maximum Score
Age	20
Medical History	30
Occupation	20
Lifestyle	20
Financial Stability	10

Risk Categories

Total Score	Category
0–30	Low Risk
31–60	Medium Risk
61–80	High Risk
81–100	Very High Risk

Business Rules

- Low-risk cases are auto-approved.
- High-risk cases require manual review.
- Very high-risk cases may be rejected.

FR-UW-007

Premium Loading

Description

Calculates additional premium for risky applicants.

Example

Risk Category	Loading
Low	0%
Medium	10%
High	25%
Very High	50%

Business Rules

- Loading percentage is configurable.
- Final premium is recalculated after loading.

FR-UW-008

Underwriting Decision

Description

Records the final underwriting decision.

Decision Types

- Approved
- Approved with Loading
- Approved with Exclusions

- Pending Medical Report
- Pending Financial Documents
- Rejected

Business Rules

- Rejection reason is mandatory.
- Only authorized underwriters can approve cases.
- Approved cases proceed to policy issuance.

Reports

The Underwriting Module provides:

- Underwriting Pending Report
- Approved Cases Report
- Rejected Cases Report
- Medical Pending Report
- High-Risk Customer Report
- Loading Report
- Underwriter Performance Report
- Product-wise Underwriting Report

Dashboard

The Underwriting Dashboard displays:

- Total Cases
- Pending Cases
- Approved Cases
- Rejected Cases

- High-Risk Customers
- Average Processing Time
- Medical Pending Cases
- Loading Applied Cases

Database Design

Table INS024 : underwriting_case

Column Name	Data Type	Constraint
underwriting_id	BIGINT	PK
quote_id	BIGINT	FK(Quote)
customer_id	BIGINT	FK(Customer)
product_id	BIGINT	FK(Insurance_Product)
underwriting_date	DATE	NOT NULL
status	ENUM('Pending','Approved','Rejected','Medical Pending','Financial Pending')	

Table INS025 : medical_underwriting

Column Name	Data Type	Constraint
medical_id	BIGINT	PK
underwriting_id	BIGINT	FK(Underwriting_Case)
height	DECIMAL(5,2)	
weight	DECIMAL(5,2)	
bmi	DECIMAL(5,2)	

Column Name	Data Type	Constraint
smoking	BOOLEAN	
alcohol	BOOLEAN	
medical_history	TEXT	
medical_report_path	VARCHAR(500)	

Table INS026 : financial_underwriting

Column Name	Data Type	Constraint
financial_id	BIGINT	PK
underwriting_id	BIGINT	FK(Underwriting_Case)
annual_income	DECIMAL(15,2)	
employer_name	VARCHAR(150)	
occupation	VARCHAR(100)	
existing_insurance	DECIMAL(15,2)	
loan_liability	DECIMAL(15,2)	

Table INS027 : risk_score

Column Name	Data Type	Constraint
risk_id	BIGINT	PK
underwriting_id	BIGINT	FK(Underwriting_Case)
medical_score	INT	
occupation_score	INT	

Column Name	Data Type	Constraint
lifestyle_score	INT	
financial_score	INT	
total_score	INT	
risk_category	ENUM('Low','Medium','High','Very High')	

Table INS028 : underwriting_decision

Column Name	Data Type	Constraint
decision_id	BIGINT	PK
underwriting_id	BIGINT	FK(Underwriting_Case)
decision	ENUM('Approved','Approved with Loading','Approved with Exclusions','Rejected')	
loading_percentage	DECIMAL(5,2)	
exclusion_details	TEXT	
rejection_reason	TEXT	
approved_by	BIGINT	FK(Employee)
decision_date	DATETIME	

Table INS029 : underwriting_audit_log

Column Name	Data Type	Constraint
audit_id	BIGINT	PK
underwriting_id	BIGINT	FK(Underwriting_Case)

Column Name	Data Type	Constraint
action	VARCHAR(100)	
performed_by	BIGINT	FK(Employee)
remarks	TEXT	
action_date	DATETIME	

RESTful API Endpoints

Underwriting APIs

Method	Endpoint	Description
POST	/api/v1/underwriting	Create Underwriting Case
GET	/api/v1/underwriting	List Cases
GET	/api/v1/underwriting/{id}	Get Underwriting Details
PUT	/api/v1/underwriting/{id}	Update Case

Medical Underwriting APIs

Method	Endpoint	Description
POST	/api/v1/underwriting/{id}/medical	Save Medical Details
GET	/api/v1/underwriting/{id}/medical	View Medical Details
PUT	/api/v1/underwriting/{id}/medical	Update Medical Details

Financial Underwriting APIs

Method	Endpoint	Description
POST	/api/v1/underwriting/{id}/financial	Save Financial Details
GET	/api/v1/underwriting/{id}/financial	View Financial Details
PUT	/api/v1/underwriting/{id}/financial	Update Financial Details

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Decision APIs

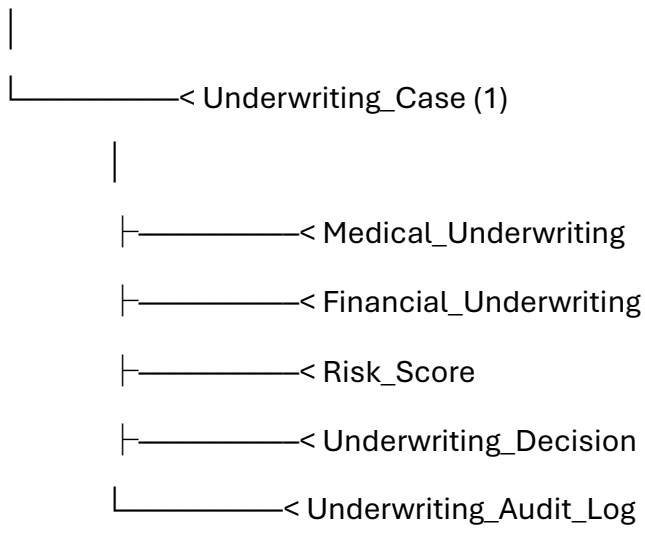
Method	Endpoint	Description
PUT	/api/v1/underwriting/{id}/approve	Approve Case
PUT	/api/v1/underwriting/{id}/reject	Reject Case
PUT	/api/v1/underwriting/{id}/loading	Apply Premium Loading
GET	/api/v1/underwriting/{id}/decision	View Decision

Angular Screens

- Underwriting Dashboard
- Underwriting Case List
- Medical Assessment
- Financial Assessment
- Risk Score Screen
- Underwriting Decision
- Premium Loading Screen
- Underwriting Audit Log

Entity Relationship

Quote (1)



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Multi-level approval workflow
- Audit logging for every underwriting action
- Secure storage of medical documents
- Server-side validation for underwriting decisions

Business Rules

- Underwriting starts only after quote generation.
- KYC verification is mandatory before underwriting.
- Medical examination is mandatory when required by product rules.
- High-risk customers require manual approval.
- Premium loading is calculated based on configurable risk rules.
- Rejected proposals cannot proceed to policy issuance.
- Every underwriting decision is permanently stored in the audit trail.

Deliverables

- Underwriting Case Management
- Medical Underwriting
- Financial Underwriting
- Occupational & Lifestyle Risk Assessment
- Risk Scoring Engine
- Premium Loading
- Underwriting Decision Workflow
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

---

7. Policy Issuance Module

Insurance Policy Administration System (PAS)

7.1 Module Overview

The **Policy Issuance Module** is responsible for converting an **approved underwriting proposal** into an active insurance policy. Once underwriting approval is received and the first premium payment is successful (or as per business rules), the system generates a unique **Policy Number**, activates the policy, generates the policy document (PDF), and notifies the customer.

The issued policy becomes the primary contract between the insurance company and the customer. It contains policy details, coverage, exclusions, premium schedule, nominee information, policy term, riders, and claim conditions.

This module integrates with:

- Customer Registration

- KYC Verification
- Product Management
- Quote Generation
- Underwriting
- Premium Payment
- Document Management
- Notification Service

7.2 Module Objectives

The Policy Issuance Module aims to:

- Generate unique policy numbers.
- Convert approved quotes into policies.
- Generate policy documents (PDF).
- Activate insurance policies.
- Configure policy effective dates.
- Manage policy endorsements.
- Store policy history.
- Notify customers after issuance.

7.3 User Roles

Role	**Permission**
Customer	View & Download Policy
Agent	Generate Policy
Operations Executive	Issue Policy
Branch Manager	Approve Special Cases

Role	Permission
Underwriter	View Issued Policy
Administrator	Full Access

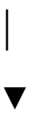
7.4 Module Components

The Policy Issuance Module consists of:

1. Policy Generation
2. Policy Number Generation
3. Policy Activation
4. Policy Document Generation
5. Coverage Details
6. Nominee Mapping
7. Rider Mapping
8. Policy Endorsement
9. Policy History
10. Notification

7.5 Policy Issuance Workflow

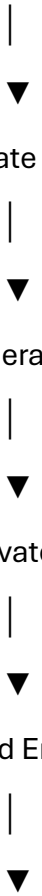
Approved Underwriting



Verify First Premium Payment



Generate Policy Number



Create Policy Record

Generate Policy PDF

Activate Policy

Send Email / SMS Notification

Policy Available in Customer Portal

Functional Requirements

FR-POL-001

Policy Generation

Description

Creates a new insurance policy from an approved quote.

Policy Details

Field	Type	Mandatory
Policy Number	Auto	Yes

Field	Type	Mandatory
Quote Number	FK	Yes
Customer ID	FK	Yes
Product ID	FK	Yes
Issue Date	Date	Yes
Effective Date	Date	Yes
Expiry Date	Date	Yes
Policy Status	Active/Inactive	Yes

Business Rules

- Policy Number is auto-generated.
- One policy can be created from one approved quote.
- Policy cannot be generated without underwriting approval.

FR-POL-002

Policy Number Generation

Description

Generates a unique policy number.

Sample Format

LIFE-2026-000012345

or

HLT-MH-2026-100245

Business Rules

- Policy Number must be unique.
- Format should be configurable.
- Duplicate policy numbers are not allowed.

FR-POL-003

Policy Activation

Description

Activates the insurance policy after successful validation.

Policy Status

- Draft
- Pending Payment
- Active
- Suspended
- Cancelled
- Expired
- Closed

Business Rules

- Policy becomes **Active** only after the required premium payment.
- Expired policies automatically change to **Expired**.

FR-POL-004

Policy Document Generation

Description

Generates the official insurance policy document.

Document Includes

- Policy Number
- Customer Details
- Nominee Details
- Coverage
- Sum Insured
- Premium
- Riders
- Terms & Conditions
- Exclusions
- Claim Procedure

Business Rules

- PDF is digitally generated.
- Every version is stored.
- Customer can download anytime.

FR-POL-005

Coverage Management

Description

Stores the coverage selected by the customer.

Coverage Details

Field	**Type**
Coverage Name	Text
Sum Insured	Decimal
Coverage Start Date	Date

Field	Type
Coverage End Date	Date

Business Rules

- Coverage cannot exceed product limits.
- Coverage starts from policy effective date.

FR-POL-006

Rider Mapping

Description

Associates optional riders with the policy.

Example Riders

- Critical Illness
- Accidental Death
- Disability
- Waiver of Premium
- Income Benefit

Business Rules

- Riders cannot be modified after policy issuance unless an endorsement is processed.

FR-POL-007

Policy Endorsement

Description

Processes policy modifications after issuance.

Supported Endorsements

- Address Change
- Nominee Change
- Coverage Increase
- Coverage Decrease
- Rider Addition
- Rider Removal
- Name Correction

Business Rules

- Endorsement history must be maintained.
- Certain endorsements require underwriting approval.

FR-POL-008

Notification Service

Description

Sends policy issuance notifications.

Notification Channels

- Email
- SMS
- WhatsApp
- Customer Portal

Business Rules

- Notification is sent immediately after successful issuance.
- Failed notifications are retried automatically.

Reports

The Policy Issuance Module provides:

- Policy Issuance Report
- Active Policy Report
- Expired Policy Report
- Cancelled Policy Report
- Product-wise Policy Report
- Agent-wise Policy Report
- Policy Endorsement Report
- Policy Renewal Due Report

Dashboard

The Policy Dashboard displays:

- Total Policies
- Active Policies
- Expired Policies
- Cancelled Policies
- Policies Issued Today
- Policies by Product
- Policies by Branch
- Pending Activations

Database Design

Table INS030 : policy

Column Name	Data Type	Constraint
policy_id	BIGINT	PK
policy_number	VARCHAR(50)	UNIQUE
quote_id	BIGINT	FK(Quote)
customer_id	BIGINT	FK(Customer)
product_id	BIGINT	FK(Insurance_Product)
issue_date	DATE	NOT NULL
effective_date	DATE	NOT NULL
expiry_date	DATE	NOT NULL
policy_status	ENUM('Draft','Pending Payment','Active','Suspended','Cancelled','Expired','Closed')	

Table INS031 : policy_coverage

Column Name	Data Type	Constraint
coverage_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
coverage_name	VARCHAR(150)	
sum_insured	DECIMAL(15,2)	
coverage_start_date	DATE	

Column Name	Data Type	Constraint
coverage_end_date	DATE	

Table INS032 : policy_rider

Column Name	Data Type	Constraint
policy_rider_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
rider_id	BIGINT	FK(Rider_Master)
rider_premium	DECIMAL(15,2)	

Table INS033 : policy_document

Column Name	Data Type	Constraint
document_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
document_name	VARCHAR(200)	
file_path	VARCHAR(500)	
version_no	INT	
generated_date	DATETIME	

Table INS034 : policy_endorsement

Column Name	Data Type	Constraint
endorsement_id	BIGINT	PK

Column Name	Data Type	Constraint
policy_id	BIGINT	FK(Policy)
endorsement_type	VARCHAR(100)	
old_value	TEXT	
new_value	TEXT	
approved_by	BIGINT	FK(Employee)
endorsement_date	DATETIME	

Table INS035 : policy_history

Column Name	Data Type	Constraint
history_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
action	VARCHAR(100)	
performed_by	BIGINT	FK(Employee)
action_date	DATETIME	
remarks	TEXT	

RESTful API Endpoints

Policy APIs

Method	Endpoint	Description
POST	/api/v1/policies	Issue New Policy
GET	/api/v1/policies	Get All Policies

Method	Endpoint	Description
GET	/api/v1/policies/{id}	Get Policy Details
PUT	/api/v1/policies/{id}	Update Policy
DELETE	/api/v1/policies/{id}	Cancel Policy

Policy Document APIs

Method	Endpoint	Description
GET	/api/v1/policies/{id}/document	Download Policy PDF
POST	/api/v1/policies/{id}/document/regenerate	Regenerate Policy PDF

Endorsement APIs

Method	Endpoint	Description
POST	/api/v1/policies/{id}/endorsements	Create Endorsement
GET	/api/v1/policies/{id}/endorsements	List Endorsements
PUT	/api/v1/endorsements/{endorsementId}/approve	Approve Endorsement
PUT	/api/v1/endorsements/{endorsementId}/reject	Reject Endorsement

---

Policy Status APIs

Method	Endpoint	Description
PUT	/api/v1/policies/{id}/activate	Activate Policy
PUT	/api/v1/policies/{id}/suspend	Suspend Policy
PUT	/api/v1/policies/{id}/cancel	Cancel Policy

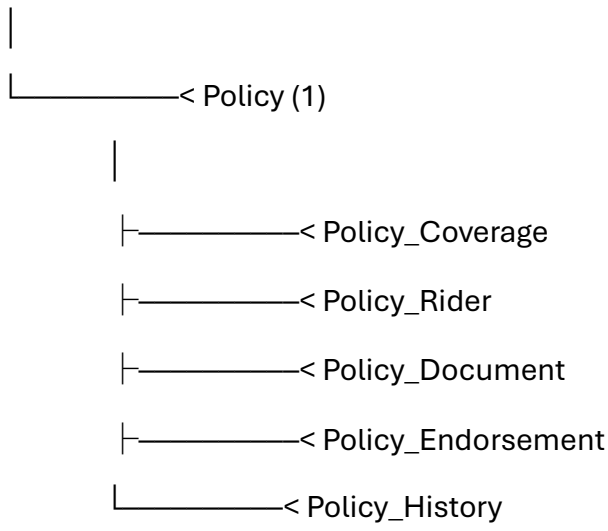
Method	Endpoint	Description
PUT	/api/v1/policies/{id}/expire	Expire Policy

Angular Screens

- Policy Dashboard
- Policy List
- Policy Details
- Policy Issuance
- Policy PDF Viewer
- Coverage Details
- Rider Details
- Endorsement Management
- Policy History

Entity Relationship

Quote (1)



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Digital policy document generation
- Secure PDF storage
- Audit logging for every policy action
- Policy version management
- API authorization using Spring Security

Business Rules

- Policy issuance is allowed only after underwriting approval.
- First premium payment is mandatory before activation (unless product rules specify otherwise).
- Policy numbers are unique and auto-generated.
- Policy documents are generated in PDF format and stored securely.
- Endorsements maintain complete version history.
- Cancelled or expired policies cannot process new claims.
- Every issuance, update, endorsement, and cancellation is recorded in the audit log.

Deliverables

- Policy Issuance
- Policy Number Generation
- Policy Activation
- Policy Document Generation
- Coverage Management
- Rider Mapping

- Policy Endorsement
- Policy History
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

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8. Premium Schedule & Premium Payment Module

Insurance Policy Administration System (PAS)

8.1 Module Overview

The **Premium Schedule & Premium Payment Module** manages the complete premium collection lifecycle after a policy is issued. It automatically generates premium schedules based on the selected payment frequency, tracks due dates, collects premium payments through multiple payment modes, applies grace periods and late payment charges, generates receipts, and updates the policy status.

The module integrates with:

- Policy Issuance Module
- Payment Gateway
- Accounting System
- Notification Service (SMS/Email)
- Customer Portal
- Agent Portal

It ensures timely premium collection and helps maintain policy continuity.

8.2 Module Objectives

The Premium Module aims to:

- Generate premium schedules.
- Calculate installment amounts.
- Accept premium payments.
- Support multiple payment methods.
- Generate payment receipts.
- Apply grace period and late fees.
- Send premium due reminders.
- Track payment history.
- Update policy status based on payments.

8.3 User Roles

Role	**Permission**
Customer	View Schedule & Pay Premium
Agent	Collect Premium
Cashier	Receive Offline Payments
Finance Officer	Verify Payments
Administrator	Configure Premium Rules

8.4 Module Components

The Premium Module consists of:

1. Premium Schedule Generation
2. Installment Management
3. Premium Payment
4. Payment Verification

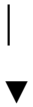
5. Payment Receipt
6. Grace Period
7. Late Fee Calculation
8. Premium Reminder
9. Payment History
10. Refund Management

8.5 Premium Collection Workflow

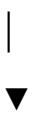
Policy Issued



Generate Premium Schedule



Premium Due Reminder

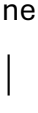


Customer Makes Payment

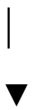
Payment Verification



Generate Receipt



Update Installment Status



Policy Remains Active

Functional Requirements

FR-PRM-001

Premium Schedule Generation

Description

Automatically generates premium installments based on policy term and payment frequency.

Schedule Details

Field	Type	Mandatory
Schedule ID	Auto	Yes
Policy ID	FK	Yes
Premium Frequency	Monthly / Quarterly / Half-Yearly / Yearly	Yes
Installment Number	Number	Yes
Due Date	Date	Yes
Premium Amount	Decimal	Yes
Status	Pending / Paid / Overdue	Yes

Business Rules

- Schedule is generated immediately after policy issuance.
- Due dates are calculated automatically.

- Installment amount depends on premium frequency.

FR-PRM-002

Installment Management

Description

Manages premium installments throughout the policy period.

Installment Status

- Pending
- Paid
- Partial Paid
- Overdue
- Cancelled

Business Rules

- Customer can view all upcoming installments.
- Paid installments cannot be modified.
- Partial payments are allowed only if configured.

FR-PRM-003

Premium Payment

Description

Allows customers to pay premium using multiple payment methods.

Supported Payment Modes

- UPI
- Credit Card
- Debit Card

- Net Banking
- Cash
- Cheque
- Demand Draft
- Wallet

Business Rules

- Online payments require payment gateway confirmation.
- Offline payments require finance verification.
- Duplicate payments are not allowed.

FR-PRM-004

Payment Verification

Description

Verifies payment status received from the payment gateway or finance team.

Payment Status

- Initiated
- Success
- Failed
- Pending
- Refunded
- Cancelled

Business Rules

- Successful payment updates installment status.
- Failed payment allows retry.

- Pending payments are verified asynchronously.

FR-PRM-005

Payment Receipt Generation

Description

Generates a digital payment receipt after successful payment.

Receipt Details

Field	Type
Receipt Number	Auto
Policy Number	Text
Customer Name	Text
Payment Date	Date
Amount Paid	Decimal
Payment Mode	Text
Transaction Reference	Text

Business Rules

- Receipt number is unique.
- Receipt is generated only for successful payments.
- PDF receipt is available for download.

FR-PRM-006

Grace Period

Description

Provides additional time after premium due date.

Example

Frequency	Grace Period
Monthly	15 Days
Quarterly	30 Days
Half-Yearly	30 Days
Yearly	30 Days

Business Rules

- Grace period depends on product configuration.
- Policy remains active during grace period.
- Claims may be subject to product-specific rules during grace.

FR-PRM-007

Late Fee Calculation

Description

Calculates penalty for overdue premium payments.

Formula

Late Fee = Overdue Days × Daily Penalty Rate

Example

Due Amount	**₹5,000**
Overdue Days	10
Penalty Per Day	₹10
Total Late Fee	₹100

Business Rules

- Late fee calculation is configurable.
- Maximum penalty amount can be configured.

FR-PRM-008

Premium Reminder

Description

Automatically sends premium reminders before and after the due date.

Notification Channels

- SMS
- Email
- WhatsApp
- Mobile App Notification

Reminder Schedule

- 15 Days Before Due Date
- 7 Days Before Due Date
- Due Date
- 3 Days After Due Date
- 7 Days After Due Date

Business Rules

- Reminder frequency is configurable.
- Notification history is maintained.

FR-PRM-009

Refund Management

Description

Processes premium refunds for cancelled or rejected policies.

Business Rules

- Refund approval is mandatory.
- Refund transaction is recorded.
- Partial refunds are supported.

Reports

The Premium Module provides:

- Premium Collection Report
- Daily Payment Report
- Monthly Collection Report
- Overdue Premium Report
- Payment Failure Report
- Refund Report
- Agent Collection Report
- Branch Collection Report

Dashboard

The Premium Dashboard displays:

- Total Premium Collected
- Today's Collection
- Pending Premium
- Overdue Premium

- Failed Transactions
- Successful Transactions
- Refund Amount
- Collection by Payment Mode

Database Design

Table INS036 : premium_schedule

Column Name	Data Type	Constraint
schedule_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
installment_no	INT	NOT NULL
due_date	DATE	NOT NULL
premium_amount	DECIMAL(15,2)	NOT NULL
late_fee	DECIMAL(15,2)	DEFAULT 0
status	ENUM('Pending','Paid','Partial Paid','Overdue','Cancelled')	

Table INS037 : premium_payment

Column Name	Data Type	Constraint
payment_id	BIGINT	PK
schedule_id	BIGINT	FK(Premium_Schedule)
policy_id	BIGINT	FK(Policy)
payment_date	DATETIME	

Column Name	Data Type	Constraint
amount_paid	DECIMAL(15,2)	
payment_mode	ENUM('UPI','Credit Card','Debit Card','Net Banking','Cash','Cheque','DD','Wallet')	
transaction_reference	VARCHAR(100)	
payment_status	ENUM('Initiated','Success','Failed','Pending','Refunded','Cancelled')	

Table INS038 : payment_receipt

Column Name	Data Type	Constraint
receipt_id	BIGINT	PK
payment_id	BIGINT	FK(Premium_Payment)
receipt_number	VARCHAR(50)	UNIQUE
receipt_date	DATETIME	
receipt_path	VARCHAR(500)	

Table INS039 : premium_reminder

Column Name	Data Type	Constraint
reminder_id	BIGINT	PK
schedule_id	BIGINT	FK(Premium_Schedule)
reminder_type	ENUM('SMS','Email','WhatsApp','Push Notification')	
reminder_date	DATETIME	

Column Name	Data Type	Constraint
delivery_status	ENUM('Pending','Sent','Failed')	

Table INS040 : premium_refund

Column Name	Data Type	Constraint
refund_id	BIGINT	PK
payment_id	BIGINT	FK(Premium_Payment)
refund_amount	DECIMAL(15,2)	
refund_date	DATETIME	
refund_status	ENUM('Pending','Approved','Completed','Rejected')	
approved_by	BIGINT	FK(Employee)

RESTful API Endpoints

Premium Schedule APIs

Method	Endpoint	Description
POST	/api/v1/premium-schedules	Generate Premium Schedule
GET	/api/v1/premium-schedules/{policyId}	Get Premium Schedule
PUT	/api/v1/premium-schedules/{scheduleId}	Update Installment

---

Premium Payment APIs

Method	Endpoint	Description
POST	/api/v1/payments	Pay Premium
GET	/api/v1/payments/{paymentId}	Get Payment Details

Method	Endpoint	Description
GET	/api/v1/payments/history/{policyId}	Payment History
POST	/api/v1/payments/verify	Verify Payment Gateway Response

Receipt APIs

Method	Endpoint	Description
GET	/api/v1/payments/{paymentId}/receipt	Download Receipt
POST	/api/v1/payments/{paymentId}/receipt/regenerate	Regenerate Receipt

---

Refund APIs

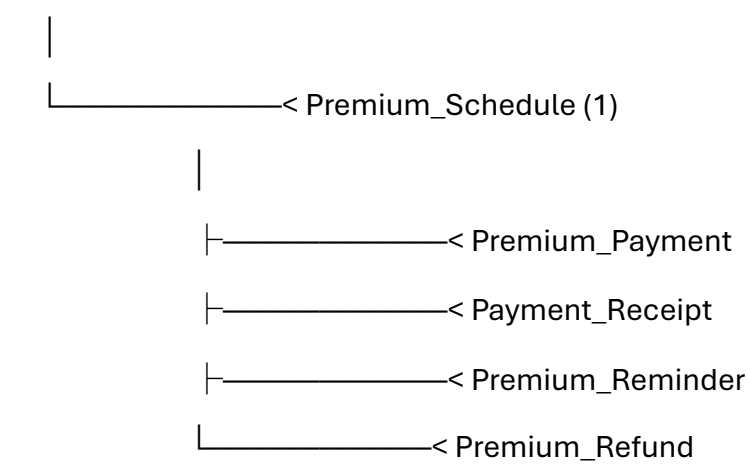
Method	Endpoint	Description
POST	/api/v1/refunds	Create Refund
PUT	/api/v1/refunds/{refundId}/approve	Approve Refund
GET	/api/v1/refunds/{refundId}	Get Refund Details

Angular Screens

- Premium Dashboard
- Premium Schedule
- Premium Payment
- Payment Gateway
- Payment Success
- Payment Failure
- Payment History
- Receipt Viewer

- Refund Management
- Premium Reminder List

Entity Relationship



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Secure payment gateway integration
- Encrypted transaction references
- Digital receipt generation
- Audit logging for every payment transaction
- PCI-DSS compliant payment handling (where applicable)

Business Rules

- Premium schedule is generated immediately after policy issuance.
- Installments are based on the selected premium frequency.
- Successful payment automatically updates installment status.

- Grace period is configurable per insurance product.
- Late fees are calculated automatically after the grace period expires.
- Refunds require approval from the Finance Officer or Administrator.
- Payment receipts are generated only for successful transactions.
- Every payment, refund, reminder, and status update is recorded in the audit log.

Deliverables

- Premium Schedule Generation
- Installment Management
- Premium Payment Processing
- Payment Verification
- Payment Receipt Generation
- Grace Period Management
- Late Fee Calculation
- Premium Reminder Service
- Refund Management
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

---

9. Policy Servicing Module

Insurance Policy Administration System (PAS)

9.1 Module Overview

The **Policy Servicing Module** manages all customer service requests after a policy has been issued. It enables policyholders to update personal information, modify policy details, request endorsements, renew policies, cancel policies, add or remove riders, and manage nominee information.

This module improves customer experience by allowing self-service through the Customer Portal while maintaining complete approval workflows and audit history.

The Policy Servicing Module integrates with:

- Policy Issuance Module
- Premium Payment Module
- Customer Portal
- Document Management
- Notification Service
- Underwriting Module
- Claims Management

9.2 Module Objectives

The Policy Servicing Module aims to:

- Manage policy endorsements.
- Update customer information.
- Change nominee details.
- Modify policy coverage.
- Add or remove riders.
- Process policy renewals.

- Process policy cancellations.
- Maintain servicing history.
- Notify customers regarding service requests.

9.3 User Roles

Role	Permission
Customer	Raise Service Request
Agent	Submit Service Request
Customer Service Executive	Process Requests
Underwriter	Approve Risk-related Changes
Branch Manager	Approve Special Requests
Administrator	Full Access

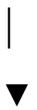
9.4 Module Components

The Policy Servicing Module consists of:

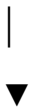
1. Policy Endorsement
2. Address Change
3. Contact Information Update
4. Nominee Management
5. Coverage Modification
6. Rider Management
7. Policy Renewal
8. Policy Cancellation
9. Service Request Tracking
10. Notification Service

9.5 Policy Servicing Workflow

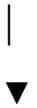
Customer Login



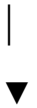
Raise Service Request



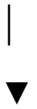
Document Verification



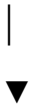
Approval Workflow



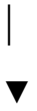
Update Policy



Generate Endorsement



Notify Customer



Service Request Closed

Functional Requirements

FR-SRV-001

Policy Endorsement

Description

Allows modification of policy details after policy issuance.

Supported Endorsements

- Address Change
- Name Correction
- Date of Birth Correction
- Coverage Increase
- Coverage Reduction
- Policy Term Change
- Sum Insured Revision

Business Rules

- Every endorsement generates a unique endorsement number.
- Some endorsements require underwriting approval.
- Previous policy information must be retained.

FR-SRV-002

Address Change

Description

Allows customers to update permanent or communication address.

Fields

Field	Type
House No	
Street	
City	
State	
Country	
PIN Code	

Business Rules

- Address proof document is mandatory.
- Address change history must be maintained.

FR-SRV-003

Contact Information Update

Description

Updates mobile number and email address.

Fields

Field	Type
Mobile Number	
Alternate Mobile	
Email Address	

Business Rules

- OTP verification required.

- Duplicate email/mobile not allowed.

FR-SRV-004

Nominee Management

Description

Allows addition or modification of nominee details.

Fields

Field	Type
Nominee Name	
Relationship	
Date of Birth	
Share Percentage	
Guardian (Minor Nominee)	

Business Rules

- Total nominee share must equal 100%.
- Guardian details mandatory for minor nominee.

FR-SRV-005

Coverage Modification

Description

Allows increasing or decreasing coverage.

Business Rules

- Coverage increase requires underwriting approval.
- Additional premium is calculated automatically.

- Coverage reduction follows company rules.

FR-SRV-006

Rider Management

Description

Allows adding or removing riders.

Supported Riders

- Critical Illness
- Personal Accident
- Disability
- Hospital Cash
- Waiver of Premium

Business Rules

- Rider eligibility depends on product.
- Premium is recalculated automatically.

FR-SRV-007

Policy Renewal

Description

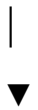
Renews insurance policies nearing expiry.

Renewal Workflow

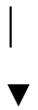
Policy Expiring



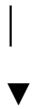
Renewal Reminder



Customer Confirmation



Premium Payment



Renew Policy

Business Rules

- Renewal reminder sent before expiry.
- Premium must be paid before renewal.
- Policy renewal updates expiry date.

FR-SRV-008

Policy Cancellation

Description

Allows cancellation of active policies.

Cancellation Reasons

- Customer Request
- Fraud Detection
- Non-payment
- Duplicate Policy
- Regulatory Requirement

Business Rules

- Refund calculation follows company policy.
- Cancellation reason is mandatory.
- Claims in progress prevent cancellation until resolved.

FR-SRV-009

Service Request Tracking

Description

Tracks servicing requests from initiation to closure.

Request Status

- Submitted
- Under Review
- Approved
- Rejected
- Completed
- Closed

Business Rules

- Every request receives a unique Service Request Number.
- Customer can track request status online.

Reports

The Policy Servicing Module provides:

- Policy Renewal Report
- Endorsement Report
- Nominee Change Report

- Rider Modification Report
- Address Change Report
- Cancellation Report
- Service Request Status Report
- Customer Activity Report

Dashboard

The Policy Servicing Dashboard displays:

- Total Service Requests
- Pending Requests
- Completed Requests
- Policy Renewals Due
- Endorsements Today
- Policy Cancellations
- Rider Changes
- Customer Satisfaction Metrics

Database Design

Table INS041 : service_request

Column Name	Data Type	Constraint
request_id	BIGINT	PK
request_number	VARCHAR(30)	UNIQUE
policy_id	BIGINT	FK(Policy)
customer_id	BIGINT	FK(Customer)

Column Name	Data Type	Constraint
request_type	VARCHAR(100)	
request_date	DATETIME	
status	ENUM('Submitted','Under Review','Approved','Rejected','Completed','Closed')	

Table INS042 : policy_endorsement

Column Name	Data Type	Constraint
endorsement_id	BIGINT	PK
request_id	BIGINT	FK(Service_Request)
endorsement_number	VARCHAR(30)	UNIQUE
endorsement_type	VARCHAR(100)	
previous_value	TEXT	
updated_value	TEXT	
approval_status	ENUM('Pending','Approved','Rejected')	

Table INS043 : nominee_details

Column Name	Data Type	Constraint
nominee_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
nominee_name	VARCHAR(100)	
relationship	VARCHAR(50)	
dob	DATE	

Column Name	Data Type	Constraint
share_percentage	DECIMAL(5,2)	
guardian_name	VARCHAR(100)	

Table INS044 : policy_renewal

Column Name	Data Type	Constraint
renewal_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
renewal_date	DATE	
new_expiry_date	DATE	
premium_amount	DECIMAL(15,2)	
renewal_status	ENUM('Pending','Paid','Renewed','Expired')	

Table INS045 : policy_cancellation

Column Name	Data Type	Constraint
cancellation_id	BIGINT	PK
policy_id	BIGINT	FK(Policy)
cancellation_date	DATE	
cancellation_reason	TEXT	
refund_amount	DECIMAL(15,2)	
approved_by	BIGINT	FK(Employee)

Table INS046 : service_request_history

Column Name	Data Type	Constraint
history_id	BIGINT	PK
request_id	BIGINT	FK(Service_Request)
action	VARCHAR(100)	
action_by	BIGINT	FK(Employee)
action_date	DATETIME	
remarks	TEXT	

RESTful API Endpoints

Service Request APIs

Method	Endpoint	Description
POST	/api/v1/service-requests	Create Service Request
GET	/api/v1/service-requests	List Requests
GET	/api/v1/service-requests/{id}	View Request
PUT	/api/v1/service-requests/{id}	Update Request

Endorsement APIs

Method	Endpoint	Description
POST	/api/v1/endorsements	Create Endorsement
PUT	/api/v1/endorsements/{id}/approve	Approve Endorsement
PUT	/api/v1/endorsements/{id}/reject	Reject Endorsement
GET	/api/v1/endorsements/{id}	View Endorsement

Renewal APIs

Method	Endpoint	Description
POST	/api/v1/policies/{id}/renew	Renew Policy
GET	/api/v1/renewals/{id}	Renewal Details

Cancellation APIs

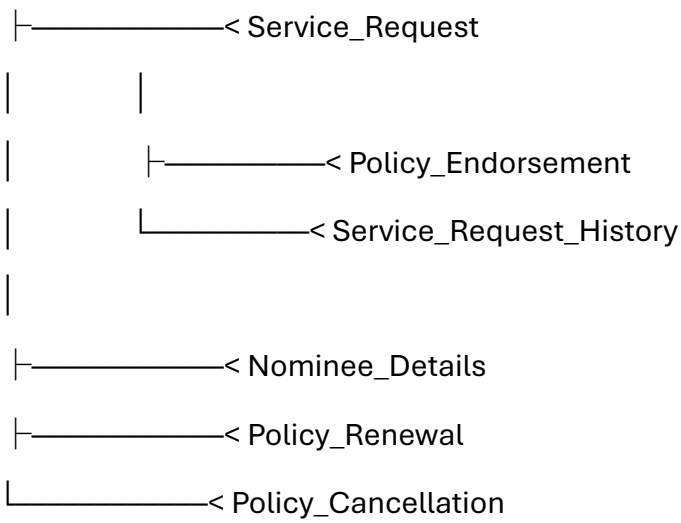
Method	Endpoint	Description
POST	/api/v1/policies/{id}/cancel	Cancel Policy
GET	/api/v1/cancellations/{id}	Cancellation Details

Angular Screens

- Policy Servicing Dashboard
- Service Request List
- Create Service Request
- Endorsement Management
- Address Change
- Nominee Management
- Rider Management
- Policy Renewal
- Policy Cancellation
- Request Tracking

Entity Relationship

Policy (1)



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- OTP verification for contact updates
- Document validation before approval
- Audit trail for every policy servicing request
- Secure API access using Spring Security

Business Rules

- Every servicing request must have a unique request number.
- Endorsements affecting risk require underwriting approval.
- Coverage increases automatically trigger premium recalculation.
- Policy renewal requires successful premium payment.
- Cancellation cannot proceed if an active claim exists.
- All changes are recorded in policy history.
- Customers receive notifications for every request status change.

Deliverables

- Policy Endorsement Management
- Address & Contact Update
- Nominee Management
- Coverage Modification
- Rider Management
- Policy Renewal
- Policy Cancellation
- Service Request Tracking
- Reports & Dashboard
- RESTful APIs
- Angular UI
- Audit Trail

---

10. Claims Management Module

Insurance Policy Administration System (PAS)

10.1 Module Overview

The **Claims Management Module** is the most critical module of the Insurance Policy Administration System (PAS). It manages the complete insurance claim lifecycle from **claim registration** to **claim settlement**. The module validates policy coverage, verifies claim documents, assigns surveyors or investigators (if required), calculates the payable claim amount, processes approvals, and completes settlement.

This module supports multiple insurance products such as:

- Health Insurance
- Motor Insurance

- Life Insurance
- Travel Insurance
- Home Insurance
- Property Insurance
- Personal Accident Insurance

The module integrates with:

- Customer Module
- Policy Module
- Premium Module
- Underwriting Module
- Document Management
- Notification Service
- Finance & Accounting

10.2 Module Objectives

The Claims Management Module aims to:

- Register insurance claims.
- Verify policy eligibility.
- Validate claim documents.
- Assign surveyors/investigators.
- Calculate claim amount.
- Approve or reject claims.
- Process claim settlements.
- Maintain claim history and audit logs.
- Notify customers throughout the claim process.

10.3 User Roles

Role	Permission
Customer	Raise Claim & Track Status
Agent	Register Claim
Claims Executive	Verify Claim
Surveyor	Submit Survey Report
Investigator	Investigate Fraud/Suspicious Claims
Claims Manager	Approve/Reject Claims
Finance Officer	Release Claim Payment
Administrator	Full Access

10.4 Module Components

The Claims Management Module consists of:

1. Claim Registration
2. Policy Verification
3. Document Verification
4. Surveyor Assignment
5. Investigation
6. Claim Assessment
7. Claim Approval
8. Claim Settlement
9. Claim History
10. Customer Notification

10.5 Claims Processing Workflow

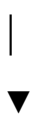
Customer Raises Claim



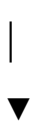
Policy Validation



Premium Status Verification



Upload Required Documents



Document Verification

Assign Surveyor (If Required)



Survey Report Submission



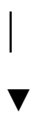
Investigation (If Required)



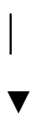
Claim Assessment



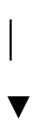
Approve / Reject Claim



Finance Payment Processing



Claim Settlement



Customer Notification

Functional Requirements

FR-CLM-001

Claim Registration

Description

Allows customers or agents to register insurance claims.

Claim Details

Field	Type	Mandatory
Claim Number	Auto	Yes
Policy Number	FK	Yes
Customer	FK	Yes
Claim Type	Dropdown	Yes

Field	Type	Mandatory
Incident Date	Date	Yes
Claim Date	Date	Yes
Description	TextArea	Yes
Estimated Loss	Decimal	Yes

Business Rules

- Policy must be active.
- Premium payments must be up to date.
- Claim Number is auto-generated.
- Incident date cannot be in the future.

FR-CLM-002

Policy Verification

Description

Validates whether the policy is eligible for claim processing.

Validation Checks

- Policy Active
- Policy Expiry Date
- Premium Status
- Coverage Available
- Waiting Period Completed
- Claim Limit

Business Rules

- Expired policies cannot raise claims.
- Lapsed policies require reinstatement.
- Waiting period rules are product specific.

FR-CLM-003

Document Verification

Description

Verifies submitted claim documents.

Sample Documents

Health Insurance

- Hospital Bills
- Medical Reports
- Discharge Summary
- Doctor Prescription

Motor Insurance

- RC Book
- Driving License
- FIR
- Vehicle Images
- Repair Estimate

Life Insurance

- Death Certificate
- Medical Certificate
- Nominee ID Proof

Business Rules

- Mandatory documents vary by claim type.
- Missing documents keep the claim in Pending status.

FR-CLM-004

Surveyor Assignment

Description

Assigns surveyors for applicable claim types.

Surveyor Responsibilities

- Site Inspection
- Damage Assessment
- Vehicle Inspection
- Property Inspection
- Report Submission

Business Rules

- Motor claims require surveyor assignment above configurable claim amounts.
- Surveyor report is mandatory before approval.

FR-CLM-005

Investigation

Description

Investigates suspicious or fraudulent claims.

Investigation Triggers

- High Claim Amount
- Multiple Claims
- Fraud Indicators

- Fake Documents
- Mismatch in Incident Details

Business Rules

- Investigation is optional.
- Investigation report is mandatory before rejection due to fraud.

FR-CLM-006

Claim Assessment

Description

Calculates the payable claim amount.

Formula

Claim Amount

-

Policy Deductible

-

Depreciation

-

Non Payable Expenses

=

Net Payable Amount

Example

Description	Amount
Claim Amount	₹2,00,000
Deductible	₹10,000
Depreciation	₹5,000
Non-Payable	₹2,000
Net Settlement	₹1,83,000

Business Rules

- Settlement cannot exceed Sum Insured.
- Deductible is product-specific.
- Depreciation rules depend on insurance type.

FR-CLM-007

Claim Approval

Description

Approves or rejects claims.

Status

- Registered
- Pending Verification

- Under Survey
- Under Investigation
- Approved
- Rejected
- Settled

Business Rules

- High-value claims require Claims Manager approval.
- Fraudulent claims are rejected with reason.

FR-CLM-008

Claim Settlement

Description

Processes payment after claim approval.

Payment Modes

- NEFT
- RTGS
- IMPS
- UPI
- Cheque

Business Rules

- Settlement only after approval.
- Bank account verification is mandatory.
- Settlement advice is generated automatically.

FR-CLM-009

Customer Notification

Description

Notifies customers during every claim stage.

Notification Channels

- SMS
- Email
- WhatsApp
- Mobile App

Notifications

- Claim Registered
- Documents Pending
- Survey Assigned
- Investigation Started
- Claim Approved
- Claim Rejected
- Payment Completed

Reports

The Claims Management Module provides:

- Daily Claim Report
- Claim Status Report
- Pending Claims Report
- Approved Claims Report
- Rejected Claims Report

- Surveyor Performance Report
- Investigation Report
- Settlement Report
- Product-wise Claim Report
- Branch-wise Claim Report

Dashboard

The Claims Dashboard displays:

- Total Claims
- Pending Claims
- Claims Under Survey
- Claims Under Investigation
- Approved Claims
- Rejected Claims
- Settled Claims
- Total Claim Amount
- Average Settlement Time
- Fraud Detection Statistics

Database Design

Table INS047 : claim_master

Column Name	Data Type	Constraint
claim_id	BIGINT	PK

Column Name	Data Type	Constraint
claim_number	VARCHAR(30)	UNIQUE
policy_id	BIGINT	FK(Policy)
customer_id	BIGINT	FK(Customer)
claim_type	VARCHAR(50)	
incident_date	DATE	
claim_date	DATE	
estimated_loss	DECIMAL(15,2)	
claim_status	ENUM('Registered','Pending Verification','Survey','Investigation','Approved','Rejected','Settled')	

Table INS048 : claim_documents

Column Name	Data Type	Constraint
document_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)
document_name	VARCHAR(150)	
file_path	VARCHAR(500)	
verification_status	ENUM('Pending','Verified','Rejected')	

Table INS049 : survey_assignment

Column Name	Data Type	Constraint
survey_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)
surveyor_id	BIGINT	FK(Employee)
assigned_date	DATETIME	
inspection_date	DATETIME	
report_path	VARCHAR(500)	
survey_status	ENUM('Assigned','Completed')	

Table INS050 : investigation

Column Name	Data Type	Constraint
investigation_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)
investigator_id	BIGINT	FK(Employee)
investigation_reason	TEXT	
report_path	VARCHAR(500)	
investigation_status	ENUM('Pending','Completed')	

Table INS051 : claim_assessment

Column Name	Data Type	Constraint
assessment_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)

Column Name	Data Type	Constraint
assessed_amount	DECIMAL(15,2)	
deductible	DECIMAL(15,2)	
depreciation	DECIMAL(15,2)	
non_payable	DECIMAL(15,2)	
net_payable	DECIMAL(15,2)	

Table INS052 : claim_settlement

Column Name	Data Type	Constraint
settlement_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)
approved_amount	DECIMAL(15,2)	
payment_mode	VARCHAR(50)	
transaction_reference	VARCHAR(100)	
settlement_date	DATETIME	
settlement_status	ENUM('Pending','Paid','Failed')	

Table INS053 : claim_history

Column Name	Data Type	Constraint
history_id	BIGINT	PK
claim_id	BIGINT	FK(Claim_Master)
action	VARCHAR(100)	

Column Name	Data Type	Constraint
performed_by	BIGINT	FK(Employee)
action_date	DATETIME	
remarks	TEXT	

RESTful API Endpoints

Claim APIs

Method	Endpoint	Description
POST	/api/v1/claims	Register New Claim
GET	/api/v1/claims	Get All Claims
GET	/api/v1/claims/{id}	Get Claim Details
PUT	/api/v1/claims/{id}	Update Claim
DELETE	/api/v1/claims/{id}	Cancel Claim (Before Processing)

Document APIs

Method	Endpoint	Description
POST	/api/v1/claims/{id}/documents	Upload Documents
GET	/api/v1/claims/{id}/documents	View Documents
PUT	/api/v1/claims/{id}/documents/verify	Verify Documents

Survey APIs

Method	Endpoint	Description
POST	/api/v1/claims/{id}/survey	Assign Surveyor
GET	/api/v1/claims/{id}/survey	View Survey Details
PUT	/api/v1/surveys/{surveyId}	Submit Survey Report

Investigation APIs

Method	Endpoint	Description
POST	/api/v1/claims/{id}/investigation	Assign Investigator
PUT	/api/v1/investigations/{id}	Submit Investigation Report

Settlement APIs

Method	Endpoint	Description
PUT	/api/v1/claims/{id}/approve	Approve Claim
PUT	/api/v1/claims/{id}/reject	Reject Claim
POST	/api/v1/claims/{id}/settlement	Process Settlement
GET	/api/v1/claims/{id}/settlement	View Settlement

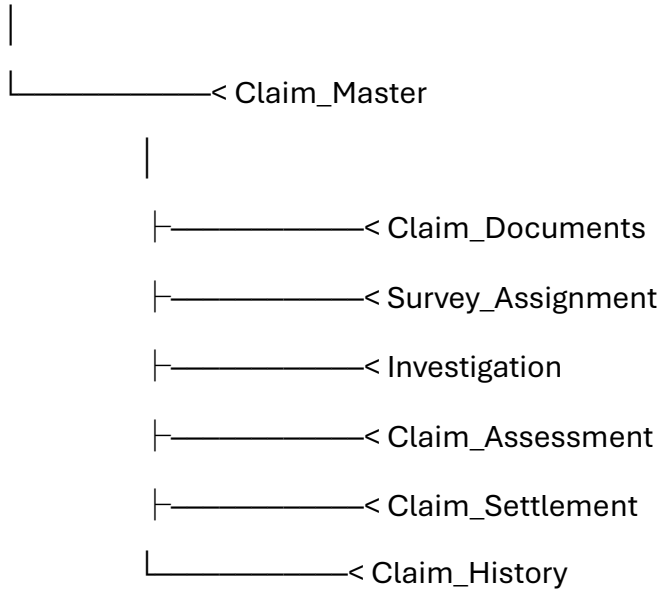
Angular Screens

- Claims Dashboard
- Claim Registration
- Claim List
- Claim Details
- Document Upload

- Survey Management
- Investigation Management
- Claim Assessment
- Claim Approval
- Claim Settlement
- Claim Tracking

Entity Relationship

Policy (1)



Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Digital document verification
- Audit trail for every claim activity
- Secure file storage
- Fraud detection rules

- Bank account verification before settlement
- Encrypted payment references

Business Rules

- Claims can only be registered for active policies.
- Premium dues must be cleared before claim registration (subject to product rules and grace period).
- Mandatory documents depend on the insurance product.
- Survey and investigation are initiated based on configurable business rules.
- Settlement amount cannot exceed the policy's sum insured.
- High-value claims require multi-level approval.
- Every claim action is recorded in the audit trail.

Deliverables

- Claim Registration
- Policy Eligibility Verification
- Document Verification
- Surveyor Assignment
- Fraud Investigation
- Claim Assessment
- Claim Approval/Rejection
- Claim Settlement
- Customer Notifications
- Reports & Dashboard
- RESTful APIs
- Angular UI

- Audit Trail

---

11. Reports & Analytics Module

Insurance Policy Administration System (PAS)

11.1 Module Overview

The **Reports & Analytics Module** provides operational, financial, and management reports for all insurance business activities. It consolidates data from Customer Registration, Policy Administration, Premium Collection, Claims, Underwriting, and Agent Management into meaningful reports, dashboards, KPIs, and analytics.

This module helps management monitor:

- Business Growth
- Premium Collection
- Policy Sales
- Claims Ratio
- Agent Performance
- Customer Statistics
- Branch Performance
- Financial Performance
- Regulatory Compliance

The module supports:

- Real-time Dashboard
- Excel Export
- PDF Export
- CSV Export
- Scheduled Reports

- Email Reports

11.2 Module Objectives

The Reports & Analytics Module aims to:

- Generate operational reports.
- Generate MIS reports.
- Generate financial reports.
- Generate regulatory reports.
- Display graphical dashboards.
- Monitor KPIs.
- Export reports.
- Schedule automatic reports.

11.3 User Roles

Role	Permission
Administrator	Full Reports Access
Branch Manager	Branch Reports
Finance Officer	Premium & Finance Reports
Claims Manager	Claim Reports
Sales Manager	Sales & Agent Reports
Underwriter	Underwriting Reports
Customer	Policy & Payment Reports (Own Data Only)

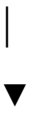
11.4 Module Components

The Reports Module consists of:

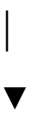
1. Customer Reports
2. Policy Reports
3. Premium Reports
4. Claims Reports
5. Underwriting Reports
6. Agent Reports
7. Financial Reports
8. Dashboard Analytics
9. Scheduled Reports
10. Export Reports

11.5 Reports Workflow

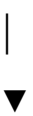
Business Transactions



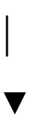
Database



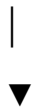
Reporting Engine



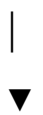
Filter & Aggregation



Charts / Tables



Dashboard



Excel / PDF / CSV Export

Functional Requirements

FR-RPT-001

Customer Reports

Description

Displays customer-related reports.

Reports

- Customer Registration Report
- Customer KYC Pending Report
- Active Customers
- Inactive Customers
- City-wise Customers
- State-wise Customers
- Product-wise Customers

Filters

- Date
- City
- State

- Product
- Agent

FR-RPT-002

Policy Reports

Description

Displays policy statistics.

Reports

- Policy Issued Report
- Active Policies
- Expired Policies
- Cancelled Policies
- Renewal Due Report
- Product-wise Policy Report
- Branch-wise Policy Report

FR-RPT-003

Premium Reports

Description

Displays premium collection reports.

Reports

- Daily Collection
- Monthly Collection
- Yearly Collection
- Pending Premium
- Overdue Premium

- Premium Frequency Report
- Refund Report

FR-RPT-004

Claims Reports

Description

Displays insurance claim reports.

Reports

- Claims Registered
- Pending Claims
- Approved Claims
- Rejected Claims
- Settled Claims
- High Value Claims
- Fraud Investigation Report

FR-RPT-005

Underwriting Reports

Description

Displays underwriting statistics.

Reports

- Pending Cases
- Approved Cases
- Rejected Cases
- Medical Pending
- High Risk Customers

- Underwriter Performance

FR-RPT-006

Agent Reports

Description

Displays agent performance.

Reports

- Agent Sales
- Premium Collection
- Claim Ratio
- Active Customers
- Commission Report
- Monthly Performance

FR-RPT-007

Financial Reports

Description

Displays financial reports.

Reports

- Premium Income
- Claim Payments
- Outstanding Premium
- Refund Report
- Branch Revenue
- Monthly Profit Summary

FR-RPT-008

Dashboard Analytics

Description

Displays graphical analytics.

Dashboard Widgets

- Total Customers
- Total Policies
- Active Policies
- Premium Collected
- Pending Premium
- Claims Registered
- Claims Settled
- Outstanding Claims
- Monthly Revenue
- Top Performing Agent

Charts

- Line Chart (Premium Trend)
- Bar Chart (Policy Sales)
- Pie Chart (Policy by Product)
- Doughnut Chart (Claim Status)
- Area Chart (Revenue Trend)
- Column Chart (Agent Performance)

FR-RPT-009

Scheduled Reports

Description

Automatically sends reports.

Schedule Options

- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

Delivery

- Email
- FTP
- Download Center

FR-RPT-010

Export Reports

Supported Formats

- PDF
- Excel
- CSV

Business Rules

- Reports respect role-based access.
- Large reports are generated asynchronously.
- Export history is maintained.

Reports Available

Customer Reports

- Customer Master
- Customer Growth

Policy Reports

- Active Policy Report
- Renewal Due

Premium Reports

- Premium Collection
- Pending Premium

Claims Reports

- Claim Settlement
- Claim Status

Financial Reports

- Revenue Report
- Expense Report

Agent Reports

- Commission Report
- Sales Report

Underwriting Reports

- Risk Analysis
- Medical Pending

Dashboard KPIs

KPI	Description
Total Customers	Registered customers
Total Policies	Issued policies
Active Policies	Running policies
Premium Collected	Total premium received
Pending Premium	Due premium
Claims Registered	Total claims
Claims Approved	Approved claims
Claims Rejected	Rejected claims
Claims Settled	Settled claims
Renewal Due	Policies nearing expiry
Revenue	Premium income
Loss Ratio	Claims ÷ Premium × 100

Database Design

Table INS054 : report_master

Column Name	Data Type	Constraint
report_id	BIGINT	PK
report_name	VARCHAR(150)	
report_type	VARCHAR(100)	
description	TEXT	

Column Name	Data Type	Constraint
created_date	DATETIME	

Table INS055 : report_schedule

Column Name	Data Type	Constraint
schedule_id	BIGINT	PK
report_id	BIGINT	FK(Report_Master)
schedule_type	ENUM('Daily','Weekly','Monthly','Quarterly','Yearly')	
next_run_date	DATETIME	
email_to	VARCHAR(250)	
status	ENUM('Active','Inactive')	

Table INS056 : report_execution

Column Name	Data Type	Constraint
execution_id	BIGINT	PK
report_id	BIGINT	FK(Report_Master)
executed_by	BIGINT	FK(Employee)
execution_date	DATETIME	
execution_status	ENUM('Success','Failed')	
file_path	VARCHAR(500)	

Table INS057 : dashboard_widget

Column Name	Data Type	Constraint
widget_id	BIGINT	PK
widget_name	VARCHAR(100)	
widget_type	VARCHAR(50)	
sequence_no	INT	
status	ENUM('Active','Inactive')	

Table INS058 : analytics_snapshot

Column Name	Data Type	Constraint
snapshot_id	BIGINT	PK
snapshot_date	DATE	
total_customers	INT	
total_policies	INT	
premium_collected	DECIMAL(15,2)	
claims_paid	DECIMAL(15,2)	
revenue	DECIMAL(15,2)	

RESTful API Endpoints

Report APIs

Method	Endpoint	Description
GET	/api/v1/reports/customers	Customer Reports
GET	/api/v1/reports/policies	Policy Reports

Method	Endpoint	Description
GET	/api/v1/reports/premiums	Premium Reports
GET	/api/v1/reports/claims	Claim Reports
GET	/api/v1/reports/underwriting	Underwriting Reports
GET	/api/v1/reports/agents	Agent Reports
GET	/api/v1/reports/financial	Financial Reports

Dashboard APIs

Method	Endpoint	Description
GET	/api/v1/dashboard/summary	Dashboard Summary
GET	/api/v1/dashboard/charts	Dashboard Charts
GET	/api/v1/dashboard/kpi	KPI Summary

Export APIs

Method	Endpoint	Description
GET	/api/v1/reports/export/pdf	Export PDF
GET	/api/v1/reports/export/excel	Export Excel
GET	/api/v1/reports/export/csv	Export CSV

Scheduled Report APIs

Method	Endpoint	Description
POST	/api/v1/reports/schedule	Create Schedule

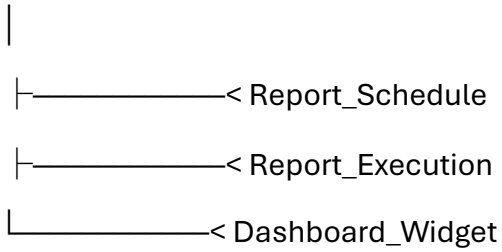
Method	Endpoint	Description
PUT	/api/v1/reports/schedule/{id}	Update Schedule
DELETE	/api/v1/reports/schedule/{id}	Delete Schedule
GET	/api/v1/reports/schedule	List Scheduled Reports

Angular Screens

- Reports Dashboard
- Customer Reports
- Policy Reports
- Premium Reports
- Claims Reports
- Underwriting Reports
- Financial Reports
- Dashboard Analytics
- Scheduled Reports
- Export Center

Entity Relationship

Report_Master (1)



Analytics_Snapshot

Security Features

- JWT Authentication
- Role-Based Access Control (RBAC)
- Report-level permissions
- Data masking for sensitive customer information
- Audit logging for report generation and exports
- Scheduled report authorization
- Secure file download

Business Rules

- Users can access only reports permitted by their role.
- Dashboard data refreshes automatically at configurable intervals.
- Scheduled reports are generated according to configured frequency.
- Exported reports are stored with execution history.
- Large reports are processed in the background.
- Sensitive customer data is masked where required by compliance policies.
- Every report generation and download is recorded in the audit log.

Deliverables

- Customer Reports
- Policy Reports
- Premium Reports
- Claims Reports
- Underwriting Reports
- Agent Reports

- Financial Reports
- MIS Dashboard
- KPI Analytics
- Scheduled Reports
- PDF / Excel / CSV Export
- RESTful APIs
- Angular Dashboards
- Audit Trail

---

12. Administration & Security Module

Insurance Policy Administration System (PAS)

12.1 Module Overview

The **Administration & Security Module** is the backbone of the Insurance Policy Administration System (PAS). It provides centralized control over users, roles, permissions, branches, departments, system configurations, master data, notifications, authentication, authorization, and audit logs.

This module ensures that only authorized users can access system resources while maintaining complete security, compliance, and traceability of all business operations.

The module integrates with every other module:

- Customer Registration
- KYC
- Product Management
- Quote Generation
- Underwriting
- Policy Issuance

- Premium Management
- Claims Management
- Reports & Analytics

12.2 Module Objectives

The Administration Module aims to:

- Manage system users.
- Manage roles and permissions.
- Configure branches and departments.
- Maintain master data.
- Configure system settings.
- Manage notification templates.
- Record audit logs.
- Secure APIs using JWT authentication.
- Control access using Role-Based Access Control (RBAC).

12.3 User Roles

Role	Permission
Super Admin	Complete System Access
Administrator	User & Configuration Management
Branch Manager	Branch Administration
Operations Manager	Operational Settings
IT Support	Technical Configuration
Auditor	Read-only Audit Access

12.4 Module Components

The Administration Module consists of:

1. User Management
2. Role Management
3. Permission Management
4. Branch Management
5. Department Management
6. Master Data Management
7. Notification Template Management
8. System Configuration
9. Audit Logs
10. Authentication & Authorization

12.5 Administration Workflow

Administrator Login



JWT Authentication

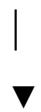


Role Verification



Access Granted

Manage Users / Masters / Configurations



Audit Log Generated

Functional Requirements

FR-ADM-001

User Management

Description

Manages all application users.

User Information

Field	Type	Mandatory
Employee ID	FK	Yes
Username	Text	Yes
Password	Encrypted	Yes
Email	Text	Yes
Mobile Number	Text	Yes
Role	FK	Yes
Branch	FK	Yes
Status	Active/Inactive	Yes

Business Rules

- Username must be unique.

- Password is stored using secure hashing.
- User account can be locked after multiple failed login attempts.
- Inactive users cannot log in.

FR-ADM-002

Role Management

Description

Defines system roles.

Sample Roles

- Super Admin
- Administrator
- Agent
- Underwriter
- Claims Executive
- Claims Manager
- Finance Officer
- Customer
- Branch Manager

Business Rules

- Each user must be assigned at least one role.
- Roles determine access to menus and APIs.

FR-ADM-003

Permission Management

Description

Controls access to system resources.

Permission Types

- Create
- Read
- Update
- Delete
- Approve
- Reject
- Export
- Print

Business Rules

- Permissions are assigned role-wise.
- Unauthorized requests return **HTTP 403 (Forbidden)**.

FR-ADM-004

Branch Management

Description

Maintains insurance branch information.

Branch Details

Field	**Type**
Branch Code	
Branch Name	
State	
City	

Field	Type
Address	
Branch Manager	

Business Rules

- Branch code must be unique.
- Users belong to one branch.

FR-ADM-005

Department Management

Departments

- Sales
- Underwriting
- Claims
- Finance
- Customer Support
- Operations
- IT
- Compliance
- HR

Business Rules

- Employees are mapped to departments.
- Reports can be filtered department-wise.

FR-ADM-006

Master Data Management

Description

Maintains lookup/master data used across the application.

Masters

- Insurance Products
- Policy Status
- Claim Status
- Payment Mode
- Gender
- Occupation
- Marital Status
- Risk Category
- Customer Type
- Country
- State
- City

Business Rules

- Master data is centrally managed.
- Deactivated master records cannot be used in new transactions.

FR-ADM-007

Notification Template Management

Description

Maintains reusable communication templates.

Notification Types

- Email
- SMS
- WhatsApp
- Push Notification

Sample Templates

- Customer Registration
- KYC Approved
- Policy Issued
- Premium Due
- Premium Paid
- Claim Registered
- Claim Approved
- Claim Rejected
- Policy Renewal Reminder

Business Rules

- Templates support dynamic placeholders (e.g., Customer Name, Policy Number).
- Template changes are version-controlled.

FR-ADM-008

System Configuration

Description

Stores configurable business settings.

Configuration Examples

- JWT Token Expiry

- Password Policy
- Session Timeout
- Maximum Login Attempts
- Grace Period Days
- Default Currency
- Date Format
- Time Zone

Business Rules

- Configuration changes are logged.
- Critical settings require Administrator privileges.

FR-ADM-009

Audit Log Management

Description

Tracks all important user activities.

Audit Information

Field	**Description**
User	
Module	
Action	
IP Address	
Login Time	
Logout Time	

Field	Description
Old Value	
New Value	

Business Rules

- Audit records cannot be edited or deleted.
- Audit logs are retained according to company policy.

FR-ADM-010

Authentication & Authorization

Description

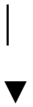
Secures the application.

Security Features

- JWT Authentication
- Spring Security
- Password Encryption (BCrypt)
- Role-Based Access Control (RBAC)
- Session Management
- Refresh Token
- API Authorization

Login Flow

User Login



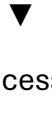
Validate Username & Password



Generate JWT Token



Return Access Token & Refresh Token



Access Protected APIs

Reports

The Administration Module provides:

- User Login Report
- Failed Login Report
- Active Users Report
- User Role Report
- Branch-wise User Report
- Department-wise User Report
- Audit Log Report
- Configuration Change Report
- Notification Delivery Report

Dashboard

The Administration Dashboard displays:

- Total Users

- Active Users
- Locked Users
- Total Roles
- Total Branches
- Login Today
- Failed Login Attempts
- Notifications Sent
- Audit Events

Database Design

Table INS059 : user_master

Column Name	Data Type	Constraint
user_id	BIGINT	PK
employee_id	BIGINT	FK(Employee)
username	VARCHAR(100)	UNIQUE
password_hash	VARCHAR(255)	NOT NULL
email	VARCHAR(150)	
mobile	VARCHAR(20)	
role_id	BIGINT	FK(Role_Master)
branch_id	BIGINT	FK(Branch_Master)
status	ENUM('Active','Inactive','Locked')	

Table INS060 : role_master

Column Name	Data Type	Constraint
role_id	BIGINT	PK
role_name	VARCHAR(100)	UNIQUE
description	VARCHAR(255)	

Table INS061 : permission_master

Column Name	Data Type	Constraint
permission_id	BIGINT	PK
permission_name	VARCHAR(100)	
module_name	VARCHAR(100)	
action_type	VARCHAR(50)	

Table INS062 : role_permission

Column Name	Data Type	Constraint
role_permission_id	BIGINT	PK
role_id	BIGINT	FK(Role_Master)
permission_id	BIGINT	FK(Permission_Master)

Table INS063 : branch_master

Column Name	Data Type	Constraint
branch_id	BIGINT	PK
branch_code	VARCHAR(20)	UNIQUE

Column Name	Data Type	Constraint
branch_name	VARCHAR(150)	
city	VARCHAR(100)	
state	VARCHAR(100)	
address	VARCHAR(255)	
manager_id	BIGINT	FK(Employee)

Table INS064 : department_master

Column Name	Data Type	Constraint
department_id	BIGINT	PK
department_name	VARCHAR(100)	UNIQUE
description	VARCHAR(255)	

Table INS065 : master_data

Column Name	Data Type	Constraint
master_id	BIGINT	PK
master_type	VARCHAR(100)	
master_value	VARCHAR(150)	
status	ENUM('Active','Inactive')	

Table INS066 : notification_template

Column Name	Data Type	Constraint
template_id	BIGINT	PK
template_name	VARCHAR(150)	
notification_type	ENUM('Email','SMS','WhatsApp','Push')	
subject	VARCHAR(255)	
body	TEXT	
version_no	INT	
status	ENUM('Active','Inactive')	

Table INS067 : system_configuration

Column Name	Data Type	Constraint
config_id	BIGINT	PK
config_key	VARCHAR(100)	UNIQUE
config_value	VARCHAR(255)	
description	VARCHAR(255)	

Table INS068 : audit_log

Column Name	Data Type	Constraint
audit_id	BIGINT	PK
user_id	BIGINT	FK(User_Master)
module_name	VARCHAR(100)	
action	VARCHAR(100)	

Column Name	Data Type	Constraint
old_value	TEXT	
new_value	TEXT	
ip_address	VARCHAR(50)	
action_time	DATETIME	

RESTful API Endpoints

Authentication APIs

Method	Endpoint	Description
POST	/api/v1/auth/login	User Login
POST	/api/v1/auth/logout	User Logout
POST	/api/v1/auth/refresh-token	Generate New Access Token
POST	/api/v1/auth/change-password	Change Password
POST	/api/v1/auth/forgot-password	Forgot Password
POST	/api/v1/auth/reset-password	Reset Password

User Management APIs

Method	Endpoint	Description
POST	/api/v1/users	Create User
GET	/api/v1/users	Get All Users
GET	/api/v1/users/{id}	Get User Details
PUT	/api/v1/users/{id}	Update User

Method	Endpoint	Description
DELETE	/api/v1/users/{id}	Delete User
PUT	/api/v1/users/{id}/lock	Lock User
PUT	/api/v1/users/{id}/unlock	Unlock User

Role & Permission APIs

Method	Endpoint	Description
POST	/api/v1/roles	Create Role
GET	/api/v1/roles	List Roles
POST	/api/v1/permissions	Create Permission
PUT	/api/v1/roles/{id}/permissions	Assign Permissions

Branch & Department APIs

Method	Endpoint	Description
POST	/api/v1/branches	Create Branch
GET	/api/v1/branches	List Branches
POST	/api/v1/departments	Create Department
GET	/api/v1/departments	List Departments

Master Data APIs

Method	Endpoint	Description
POST	/api/v1/masters	Create Master Data

Method	Endpoint	Description
GET	/api/v1/masters	Get Master Data
PUT	/api/v1/masters/{id}	Update Master Data
DELETE	/api/v1/masters/{id}	Delete Master Data

Notification Template APIs

Method	Endpoint	Description
POST	/api/v1/notification-templates	Create Template
GET	/api/v1/notification-templates	Get Templates
PUT	/api/v1/notification-templates/{id}	Update Template
DELETE	/api/v1/notification-templates/{id}	Delete Template

Audit APIs

Method	Endpoint	Description
GET	/api/v1/audit-logs	Get Audit Logs
GET	/api/v1/audit-logs/{id}	View Audit Details

Angular Screens

Authentication

- Login
- Forgot Password
- Reset Password
- Change Password

User Administration

- User List
- Add User
- Edit User
- User Profile
- Lock/Unlock User

Security

- Role Management
- Permission Management

Administration

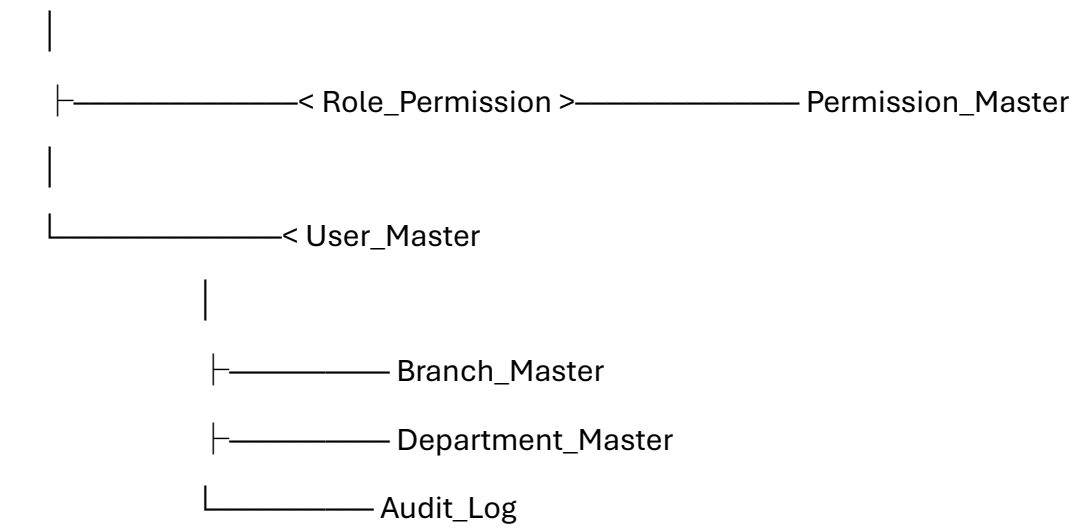
- Branch Management
- Department Management
- Master Data Management
- Notification Template Management
- System Configuration

Audit

- Audit Log Dashboard
- Login History
- User Activity History

Entity Relationship Diagram

Role_Master (1)



Master_Data

Notification_Template

System_Configuration

Security Features

- JWT Authentication
- Spring Security Integration
- BCrypt Password Encryption
- Refresh Token Support
- Role-Based Access Control (RBAC)
- Method-Level Security (@PreAuthorize)
- HTTPS Communication
- CORS Configuration
- CSRF Protection (for applicable flows)
- SQL Injection Protection using JPA/Hibernate
- XSS Input Validation
- Secure File Upload Validation

- Audit Trail for all critical operations
- Session Timeout & Auto Logout
- Password Expiry Policy
- Account Lock after configurable failed login attempts

Business Rules

- Every user must belong to at least one role.
- Access to menus, APIs, and reports is controlled through RBAC.
- Passwords are never stored in plain text.
- Audit logs are immutable and retained as per organizational policy.
- Master data changes are immediately reflected across dependent modules.
- Critical configuration changes require Administrator privileges.
- Locked or inactive users cannot authenticate.
- Notification templates support dynamic placeholders and version history.

Deliverables

- Authentication & Authorization (JWT + Spring Security)
- User Management
- Role & Permission Management
- Branch Management
- Department Management
- Master Data Management
- Notification Template Management
- System Configuration
- Audit Logging
- Security Controls

- RESTful APIs
- Angular Administration Screens
- Complete RBAC Implementation

Final Insurance Project Summary

Technology Stack

Layer	**Technology**
Language	Core Java, JDK 8
Backend	Spring Boot, Spring MVC
ORM	JPA, Hibernate, HQL, JPQL
API	RESTful APIs
Database	MySQL
Security	Spring Security + JWT
Frontend	Angular
UI	Bootstrap 5, HTML5, CSS3
Build Tool	Maven
IDE	IntelliJ IDEA / Eclipse / STS
API Testing	Postman
Version Control	Git, GitHub
Documentation	Swagger/OpenAPI

Complete Business Flow

Customer Registration

↓

KYC Verification

↓

Product Selection

↓

Quote Generation

↓

Underwriting

↓

Policy Issuance

↓

Premium Schedule Creation

↓

Premium Payment

↓

Policy Active

↓

Policy Servicing (Renewal/Endorsement)

↓

Claim Registration

↓

Survey / Investigation

↓

Claim Assessment

↓

Claim Approval / Rejection

↓

Claim Settlement

↓

Reports & Analytics

↓

Administration & Security